

Classification and Morphology of External Galaxies.

By

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With 7 Figures and 11 Plates.

Introduction.

The earliest systems of classification of “nebulae” by W. HERSCHEL and others have been described by H. D. CURTIS in his article of the “Handbuch der Astrophysik” [A]¹. The *spiral* form of some nebulae was first detected visually by Lord ROSSE and his assistants between 1845 and 1850² and the appellation *elliptical* first used in a purely descriptive way was applied to non-spiral tub presumably external stellar systems by S. ALEXANDER in 1852³.

The great abundance of the spiral type was brought to light by the photographic surveys of the end of the XIX-th century. The spiral structure of the Andromeda nebula was detected photographically in 1888 by I. ROBERTS. These surveys also revealed the great abundance of smaller objects of the smooth, structureless elliptical type and the very elongated objects described as “spindles”. In 1918 CURTIS [9] isolated a special type of spiral, the so-called “ Φ ” type, characterized by a diametral bar across the nucleus and a general ring-like structure; he also identified the “spindle” nebulae with a longitudinal dark lane as edgewise spirals with peripheral dark matter [10].

An early alphabetical classification of apparent shapes, introduced by M. WOLF⁴, has been extensively used at Heidelberg by REINMUTH [62] and at Lund by LUNDMARK [44], HOLMBERG, REIZ, DANVER and others⁵. Its relation to the standard classification has been given by SHAPLEY and Miss AMES [68]. For faint nebulae which show little or no structure, a descriptive classification, based on “concentration” and “elongation”, introduced by SHAPLEY⁶ in 1929, has been in use at Harvard for some years; it bears little relation to the standard classification.

Rather similar systems of morphological classification were introduced, about 30 years ago, by HUBBLE [25] and by LUNDMARK [43] with three main types, viz. elliptical, spiral (normal and barred) and irregular. This scheme, extensively used and later further developed by HUBBLE, has been accepted as standard up to the present time.

¹ The main references are listed at the end of the chapter “General physical properties of external galaxies”, p. 311.

² Earl of ROSSE: Phil. Trans. Roy. Soc. Lond. **1850**; **1861**.

³ S. ALEXANDER: Astronom. J. **2**, 95.

⁴ M. WOLF: Veröff. Königstuhl-Heidelberg **3**, Nr. 5 (Plate V).

⁵ See Ann. Lund Obs. **6** (1937); **9** (1941); **10** (1942).

⁶ H. SHAPLEY: Harvard Bull. No. 849.

I. Classification.

1. Standard classification. This is the Mt. Wilson classification as used by HUBBLE between 1925 and 1935; it has been so often described [*A, B, D*] that only a short summary will suffice here. It is illustrated by HUBBLE's well known "tuning-fork" diagram (Fig. 1).

a) Description of types.

α) *Elliptical nebulae* (*E*), range from circular or globular objects, such as NGC 3379, to elongated, lenticular objects, such as NGC 3115. As a rule they show no structural details, besides a small, bright and strongly condensed nucleus around which the textureless nebulosity decreases smoothly outwards in all directions to an indefinite edge where it fades into the general luminosity of the night sky.

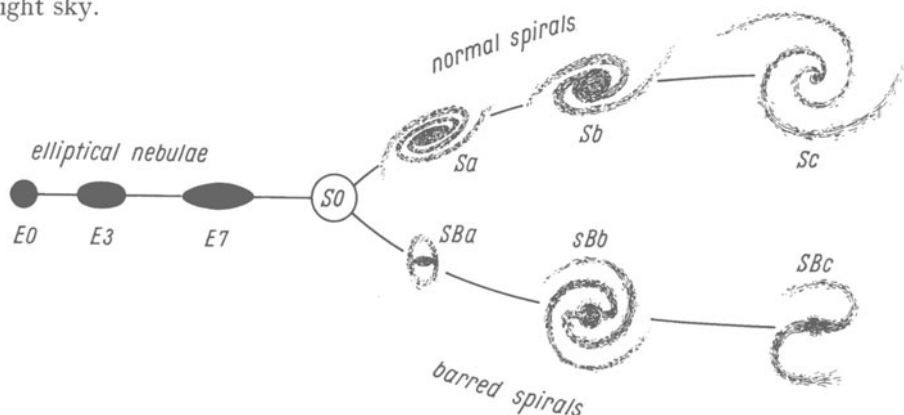


Fig. 1. Standard classification: HUBBLE's tuning-fork diagram (1925).

Sub-types are defined by the index $n = 10(1 - b/a)$, if a, b are the apparent major and minor axes measured on photographs. The most strongly elongated objects, type *E7*, such as NGC 3115, depart notably from a geometrical "elliptical" shape, being pointed near the ends of the major axis.

β) *Normal spirals* (*S*) show the characteristic spiral arms when seen pole-on and, as a rule, a "spindle" shape with heavy absorption lanes of dark matter when seen edge-on. In the normal spirals the arms emerge tangentially from a bright central nucleus at opposite points on its indefinite edge and vanish after about one complete turn of the best fitting logarithmic spiral [12]; (cf. Sect. 6). In the more regular or classical examples only two main arms, very nearly symmetrical with respect to the nucleus, are present. In most cases, however, additional or secondary arms may exist and the spiral pattern is often far from regular.

Sub-types, noted a, b, c , are defined by the relative importance of the nucleus (decreasing from a to c) and the degree of unwinding and resolution of the arms (increasing from a to c). According to HUBBLE [25]: "the arms appear to build up at the expense of the nuclear regions and unwind as they grow; in the end the arms are wide open and the nuclei inconspicuous. Early in the series the arms begin to break up into condensations, the resolution commencing in the outer regions and working inwards until in the final stages it reaches the nucleus itself" ([5], p. 326). The resolution referred to is into blue supergiants and emission objects characteristic of a Type I population. The gradual decrease of the axial ratio nucleus/spiral arms is best seen in edgewise systems (see Plate VIII),

while face-on systems show more clearly the increasing resolution and irregularity from “early” types (*Sa*) to “late” types (*Sc*) (Plate V)¹.

Intermediate types: HUBBLE also introduced the notion of “lateral” extension or width of the classification sequence in its intermediate section, giving M 81 with “large nuclear region and thin, rather open arms” and M 94 “having smaller nuclear region with closely coiled arms” as extreme cases. This distinction has been little used in practice.

Of more importance was the recognition of objects intermediate between normal and barred spirals, such as M 83 and M 61 which have been classified alternatively as *Sc* or *SBc*. Their intermediate characteristics first noticed by HUBBLE ([*B*], p. 46) and LUNDMARK [43], [44] have been discussed by LINDBLAD and LANGEARTEL [41]². They more or less fill the gap between the two branches of the tuning-fork diagram.

γ) *Barred spirals (SB)*, include the “pin-wheel” or “ Φ -type” first described by CURTIS [9]. In it a very bright central nucleus is crossed diametrically by a bar at the extremities of which spiral arms start at right angles (in “late” sub-types) or tangentially from the rim of a continuous ring of which the bar is a diameter (in “early” sub-types). Additional or secondary arms may exist, but as a rule the symmetry of the pattern is more regular than in normal spirals (see Plate VI).

Sub-types, noted *a, b, c*, are defined as for the normal spirals by the relative size of the nucleus and the degree of resolution and opening of the spiral structure. In HUBBLE’s original system the ring, closed in the *SBa* and *SBb* sub-types, opens at *SBc*, producing the aspect sometimes described as “S-shaped” spirals. *SBa* objects observed under various angles give rise to singular “Saturn-like” shapes (Plate X).

δ) *Irregulars (I)*, were described originally by HUBBLE [25] as a class of objects “lacking both dominating nuclei and rotational symmetry” and of which “the Magellanic Clouds are the most conspicuous examples” ([25], p. 328). These were termed more specifically “Magellanic nebulae” by LUNDMARK [43], [44]. However, the class was broadened by HUBBLE to include peculiar or chaotic objects which “do not find a place in the sequence of classification” since “the remaining irregulars might be arbitrarily placed in the regular sequence as highly peculiar objects, rather than in a separate class... Others, such as M 82, are merely nondescript” ([*B*], p. 47).

In fact the symbol *I* has often been used as almost equivalent to the subscript *p* for “peculiar”; such an extension of the notation is both confusing and unwarranted. As a result the relation of “irregulars” to other sections of the classification sequence was not clear, some of them being clearly related to late-type spirals and others to early-type spirals

b) Frequency of types.

From 600 bright galaxies in the Lick and Mt. Wilson plate collections HUBBLE [25] found the following apparent relative frequencies:

Type	<i>E</i>	<i>Sa, SBa</i>	<i>Sb, SBb</i>	<i>Sc, SBc</i>	<i>I</i>
Frequency	17%	19%	25%	36%	2.5%

¹ In accordance with establish custom, the words “early” and “late” are used in connection with the position in the spiral sequence but have no temporal connotation.

² See also in [38] a discussion by LINDBLAD of the barred spiral characteristics in the Andromeda nebula.

Hence the spirals constituted about 80% of the sample, with $(S + SB)/E = 4$ or 5 and, further, a ratio $S/SB = 2$ or 3.

However, SHAPLEY and AMES¹ found that in the Coma-Virgo region for $m < 12$ spirals comprise only 46% of the population and for $m < 14$, $S = 48\%$, $E = 47\%$, $I = 5\%$.

These early statistics, although not necessarily inconsistent, were affected by selection effects arising from clustering and different interpretations of some nebular types not included in the standard classification. Frequencies based on revised types are given in Sect. 3 c.

2. Revisions and additions. About 1935 Hubble undertook a systematic morphological study of the approximately 1000 brighter galaxies listed in the Shapley-Ames Catalogue, north of -30° declination, with a view to refining his original classification scheme. This work begun with the 100-inch reflector and later continued with the 200-inch reflector was practically completed at the time of HUBBLE's death in 1953; his notes collected and organized by SANDAGE will be published in the near future with a collection of about 175 photographs illustrating the morphological features characterizing the various galaxy types. Through the courtesy of Dr. A. R. SANDAGE the main revisions introduced by HUBBLE to his system can be briefly described here.

a) The most important addition was the introduction of the *SO* and *SBO* types regarded as transition stages between the ellipticals and spirals at the branching off point of the tuning-fork.

SO objects have the smooth appearance of ellipticals, but a luminosity distribution more like that of spirals, although no spiral arms are visible. They are characterized by a sharp, bright nucleus in the centre of a more or less uniform disc or "lens" having a rather sharp outer rim and surrounded by a faint, diffuse "envelope" with indefinite boundaries; the diameter of the lens in which dark crescents or rings of obscuring matter are often observed is usually about one-third of that of the envelope (see Plate III).

There is apparently a continuous transition from "late" ellipticals, such as NGC 4270, 4958, etc. and "early" *SO*'s such as NGC 3630, 4564, etc., then through various stages of the *SO* sequence to the "earliest" regular spirals.

HUBBLE distinguished two groups of *SO* objects:

SO(1): smooth lens and envelope; early examples are NGC 1201, 1332, and late examples NGC 3065, 4684.

SO(2): some structure in the envelope in the form of a dark zone and ring; examples are NGC 4459 and 4111, the latter seen edgewise.

In edgewise objects the presence of rings manifests itself by the appearance of "ansae" simulating Saturn's ring, such as in NGC 4215, 7332, etc. (Plate IX). The transition stage *SO/a* between *SO* and *Sa* shows incipient spiral structure in the envelope.

SBO objects are characterized by a bar through the central lens, sometimes broad and hazy, sometimes narrow and sharp; the envelope may form faint outer rings, sometimes conspicuous, sometimes vague or imperceptible (Plate IV). HUBBLE distinguished three groups of *SBO* objects:

SBO(1): a bright lens, with broad, hazy bar and no ring, surrounded by a larger, fainter envelope, e.g. as in NGC 3384, 4262, etc., some have circular envelopes, e.g. NGC 4203.

SBO(2): a broad, weak bar across a primary ring, with faint outer secondary rings, e.g. NGC 2859.

¹ H. SHAPLEY and A. AMES: Harvard Bull. 1926, No. 838, 3; 1930, No. 876, 39.

SBO(3): well developed bar and ring pattern, with the bar stronger than the ring, e.g. NGC 4643, 5101.

b) Many of the objects originally classified as *SBa* were then reclassified as *SBO* and the *SBa* sub-division had to be redefined as follows:

SBa: smooth bar and lens, with poorly developed, closely coiled arms in envelope and either massive and structureless or filamentary and partially resolved.

HUBBLE further distinguished two or three groups of barred spirals, one in which the arms extend from the rim of a ring crossed by a bar, e.g. NGC 2217, 5566, 5701 (*SBa*), NGC 4999, 5950 (*SBb*), and one in which the arms start at the ends of the bar without ring, e.g. NGC 2798, 4290, 7743 (*SBa*), NGC 1300, 5430, 6951 (*SBb*). In another group still the ring is formed of closely coiled filamentary arms, e.g. NGC 3185, 4037, 4385, 4389. This distinction remains well marked at the *SBb* stage, showing well developed, partially resolved arms going through more than one revolution, but there is no spiral structure in the lens; primary rings consist of spiral arms, secondary rings are seldom found. It was not followed up into the *SBc* sub-division, characterized by wide open, well resolved spiral arms with absorption lanes and spiral structure in the lens, but no ring pattern.

SANDAGE has also recognized two types of normal spirals, one in which the arms start at the rim of a ring structure, the other in which they start from a central nucleus. Some objects of the ringed type had previously been described by SHAPLEY and PARASKEVOPOULOS¹ as "plate spirals" and an investigation of ring structures was made by RANDERS [57].

There are further differences in the *multiplicity* of the spiral pattern, some objects showing only two main regular arms, others having a great many tightly coiled whorls. REYNOLDS had already pointed out² that some spirals have "massive" arms, e.g. M 31, M 33, while others have "filamentary" arms, e.g. M 81, M 101. The importance of this distinction was acknowledged by HUBBLE [B] but could not be incorporated in the classification.

c) An extension of the normal spiral sequence beyond the stage *Sc* was proposed by SHAPLEY¹ who used the notation *Sd* for objects such as NGC 7793 (Plate V, 20), showing a very small, bright nucleus and many knotty irregular spiral arms. This notation could also, and perhaps more appropriately, be applied to highly disorganized and complex spirals of low surface brightness, such as NGC 4395—4401 (see Plate VII).

A parallel extension of the barred spiral sequence beyond the stage *SBc* was introduced by de VAUCOULEURS³ through the recognition of spiral structure in the Magellanic Clouds and objects of similar type, such as NGC 1313 (Plate VI, 32), 4027, 4618, etc., which may be noted as *SBd* or *SBm*.

Still "later" stages extending the spiral sequence into the irregular types may be represented by objects such as IC 2574 or NGC 2366, IC 4662, etc. These objects are characterized by low surface brightness, high degree of resolution and sometimes outstanding emission nebulosities similar to 30 Doradus in the Large Cloud. They always show an abundance of blue supergiants and strong emission lines in their spectrum.

¹ H. SHAPLEY and J. S. PARASKEVOPOULOS: Proc. Nat. Acad. Sci. U.S.A. **26**, 31—36 (1940) = Harvard Rep. 184.

² J. H. REYNOLDS: Observatory **50**, 185—189 (1927); with comments by HUBBLE, Observatory **50**, 276—281 and by REYNOLDS, Observatory **50**, 308.

³ G. DE VAUCOULEURS: Observatory **74**, 23—31 (1954). — Astronom. J. **60**, 126—140, 219—230 (1955).

An important characteristic of the group of irregulars related to the magellanic type $I(m)$ is their small diameter and low luminosity which marks them as dwarf galaxies. Typical objects of this group are NGC 6822, IC 1613, the Sextans system, the Wolf-Lundmark-Melotte nebula, etc.¹ (Plate VII).

In the poorest and smallest of them emission objects may be few or absent, but when their distance is small enough they are always well resolved into blue supergiants and giants.

d) The existence of dwarf ellipticals (dE) of very low surface brightness was first brought to notice in 1938 through the discovery of the Fornax and Sculptor systems by SHAPLEY². These are close enough to be resolved on blue plates at $m \approx 18$; except for their very low density they seem to share all other population characteristics of normal ellipticals. Because of their very low surface brightness such systems are difficult to detect and might be much more abundant in space than their belated discovery suggests; nevertheless a special search for them on Harvard plates failed to disclose other examples. A few more "Sculptor type" systems have, however, been found since 1949 with the 48-inch Palomar Schmidt³.

Dwarf galaxies of a possibly related type have been investigated in 1952 by REAVES [58] on plates of the Virgo cluster taken by C. D. SHANE with the Lick 20-inch astrograph. These objects of which the brightest example in the Virgo cluster is IC 3475 have the same smooth, circular or little elongated appearance as the dwarf ellipticals and likewise show little central condensation. HUBBLE and REAVES suggested that they may be related to a group of dwarf spirals of low surface brightness exemplified by NGC 3299 because their colour was at first thought to be blueish; however, recent observations indicate them to be reddish and this strengthens the similarity with the Sculptor-type systems.

e) After all such additional types or variants have been weaned out, there remains a hard core of "irregular" or "peculiar" objects which do not seem to fit in any of the recognized types. One group of such objects consists of strongly interacting or colliding systems such as NGC 1275, 4038—39, 5128, etc.⁴ which can often be identified by their distorted structure, abnormal spectrum and strong radio-emission; these systems are discussed in the article by B. Y. MILLS. Double or multiple systems showing moderate interaction in the form of connecting links, distorted spiral arms or irregular filamentary extensions are discussed by F. ZWICKY. Only isolated galaxies or weakly interacting systems are considered in the present article.

Even among them, however, there remains a small number of objects not clearly assignable to any of the previous types; one possible group, exemplified by NGC 3034, NGC 3077, etc. is characterized by an early-type spectrum (A, F) contrasting with a reddish colour ($C \approx +0.8$), irregular absorption patches and filaments and a smooth, unresolved nebulous structure indicating an absence of blue supergiants and of discrete emission nebulosities (see Plate XI). Another possible group, exemplified by NGC 5253⁵ is characterized by fairly strong

¹ See E. HUBBLE: *Astrophys. Journ.* **62**, 409—433 (1925) = M.W.C. 304. — W. BAADE: *Astronom. Nachr.* **234**, 407 (1929). — F. ZWICKY: *Phys. Rev.* **58**, 478 (1940). — K. LUND-MARK: *V.J.S.* **68**, 382 (1933) = *Lund. Medd.* (I), Nr. 135.

² H. SHAPLEY: *Bull. Harvard Coll. Obs.* **1938**, No. 908.

³ R. G. HARRINGTON and A. G. WILSON: *Proc. Astr. Soc. Pacific* **62**, 118—120 (1950). — A. G. WILSON: *Proc. Astr. Soc. Pacific* **67**, 27—29 (1955).

⁴ See W. BAADE and R. MINKOWSKI: *Astrophys. Journ.* **119**, 215—231 (1954). — F. ZWICKY: *Ergebn. exakt. Naturw.* **29**, 344—385 (1956).

⁵ D. S. EVANS: *Observatory* **72**, 164—166 (1952).

emission lines and a complex structure unlike that of the magellanic irregulars or late-type spirals. Still other puzzling objects such as MAYALL's nebula¹, also show strong emission lines but have a fairly smooth structure².

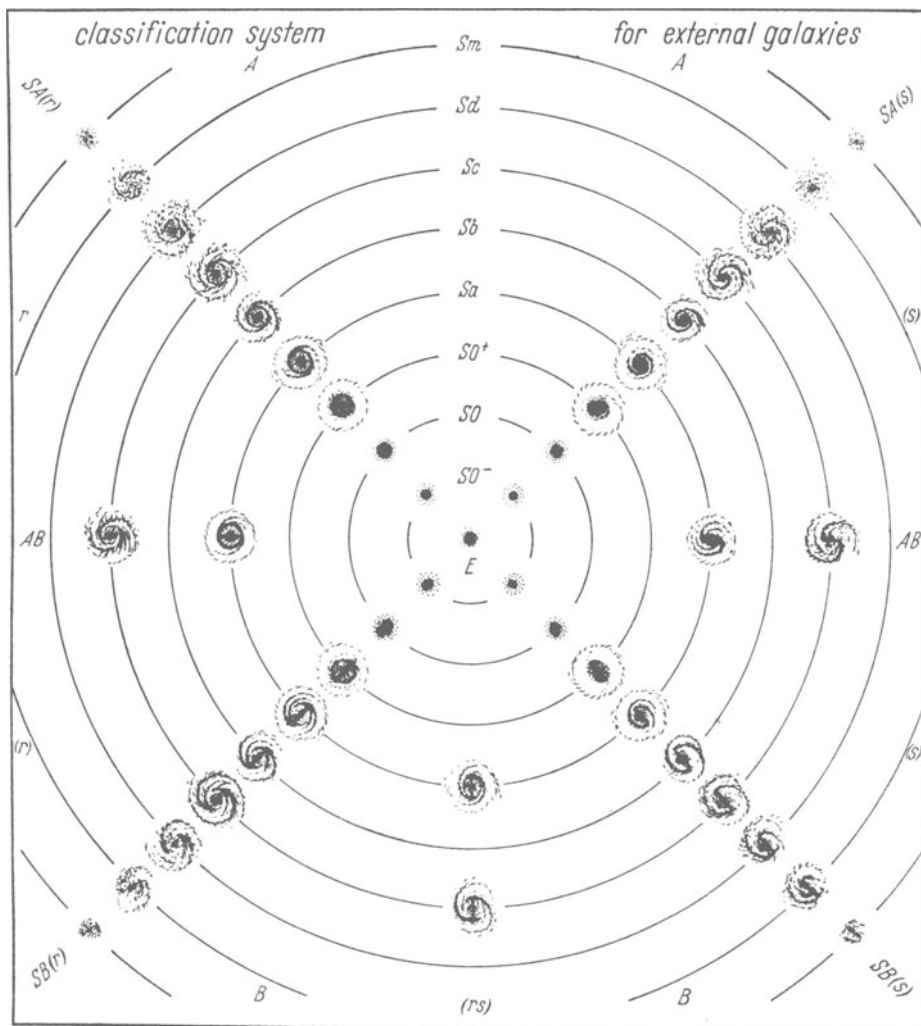


Fig. 2. Revised classification: a plane projection of the classification volume. Compare with Fig. 3. The ordinary spirals SA are in the upper half of the figure, the barred spirals SB in the lower half. The ring types (r) are to the left, the spiral types (s) to the right. Ellipticals and lenticulars are near the center, magellanic irregulars near the rim. The main stages of the classification sequence from E to Im through SO^- , SO , SO^+ , Sa , Sb , Sc , Sd , Sm are illustrated, approximately on the same scale, along each of the four main morphological series $SA(r)$, $SA(s)$, $SB(s)$, $SB(r)$. A few mixed or "intermediate" types SAB and $S(rs)$ are shown along the horizontal and vertical diameters respectively. Structures predominantly of Type I population are dashed, of Type II dotted. This classification scheme, as used in the Mount Stromlo survey of southern galaxies, is superseded by the slightly revised and improved system illustrated in Fig. 3 and Plates I to XI.

3. Revised classification. In the course of a survey of bright southern galaxies with the 30-inch Reynolds reflector at Mount Stromlo [80] a classification and

¹ R. T. SMITH: Proc. Astr. Soc. Pacific **53**, 187 (1941); also Astrophys. Journ. **119**, 225 (Fig. 14) (1954).

² Some early-type spirals are known to show abnormally wide emission lines in the spectra of their nuclei, although the origin of such high random velocities (≈ 5000 km/sec) is not known, this need not be considered as an indication of a distinct type, but merely of a peculiarity.

notation system has been developed to include in a consistent scheme all or most of the recent revisions and additions to the standard classification. The classification follows as closely as possible the revised Mt. Wilson-Palomar scheme, but includes the additional types and sub-types suggested by the Harvard, Lick and Mt. Stromlo work; the notation is based on the original Mt. Wilson system, but has been modified and supplemented as required by the intro-

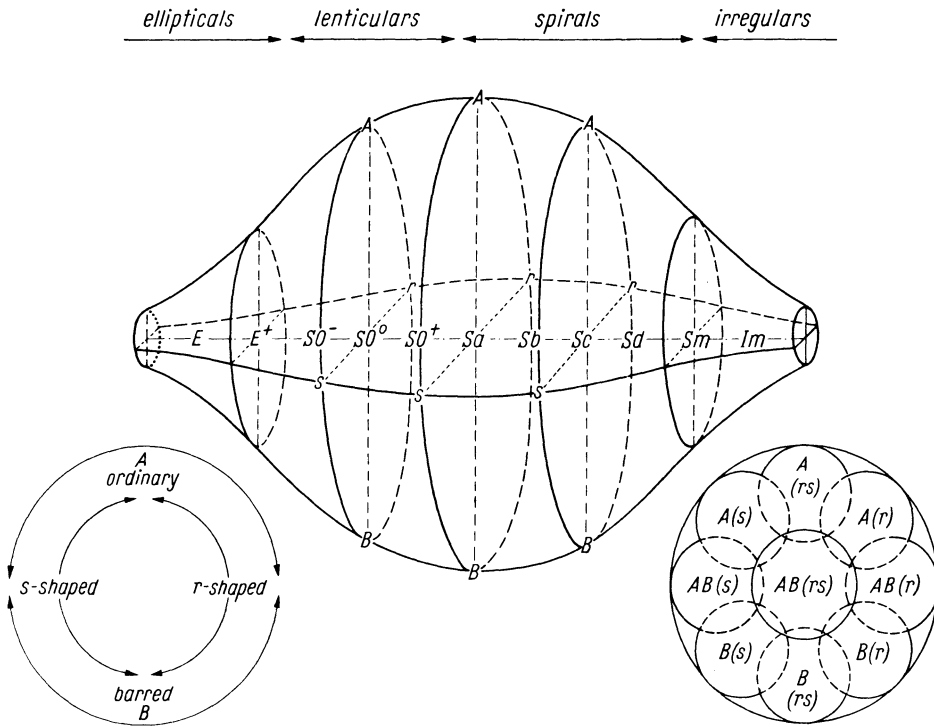


Fig. 3. A 3-dimensional representation of the revised classification scheme and notation system. From left to right are the four main classes: ellipticals E , lenticulars SO , spirals S irregulars I . Above are the ordinary families SA , below the barred families SB ; on the near side are the S -shaped varieties $S(s)$, on the far side the ringed varieties $S(r)$. The shape of the volume indicates that the separation between the various sequences $SA(s)$, $SA(r)$, $SB(r)$, $SB(s)$ is greatest at the transition stage SO/a between lenticulars and spirals and vanishes at E and Im . A central cross-section of the classification volume illustrates the relative location of the main types and the notation system. There is a continuous transition of mixed types between the main families and varieties across the classification volume and between stages along each sequence; each point in the classification volume represents potentially a possible combination of morphological characteristics. For classification purposes this infinite continuum of types is represented by a finite number of discrete "cells." Compare with Plates I and II.

duction of new types and sub-types and so as to permit a finer description of structural features. Classification and notation are illustrated in Fig. 2 which may be regarded as an extension of HUBBLE's tuning-fork diagram.

Actually Fig. 2 may be considered as a plane projection of a three dimensional representation, the idea of which was first conceived during conversations with Dr. SANDAGE in August 1955. It is schematically illustrated in Fig. 3 which may be consulted in conjunction with Fig. 2 to follow the detailed description of the adopted classification and notation (compare also with Plates I and II). The three-dimensional classification appears necessary to represent objects of mixed characteristics in correct relation to the main sequences as well as the progressive divergence of the various sequences from E to SO/a through the SO stages and their ulterior convergence from SO/a to I through the S stages.

a) Isolated galaxies.

The present classification scheme and notation system rest on the following principles:

Four *classes* are retained: ellipticals E , lenticulars SO , spirals S , irregulars I , which coincide with the main divisions introduced by HUBBLE.

The two *families* of lenticulars and spirals, originally denoted "normal" S and "barred" SB by HUBBLE, are now designated "ordinary" SA and "barred" SB , so as to permit the use of the compound symbol SAB for "intermediate" objects of mixed characteristics. The symbol S alone is used when a spiral object cannot be more accurately classified as either SA or SB because of poor resolution, unfavorable tilt, etc. The "ordinary" spirals are not more "normal" than those of the "barred" family which are at least as common (cf. Sect. 3c).

In the new class of lenticulars the two families are denoted SAO and SBO , depending on the absence or presence of a bar structure across the central lens; "intermediate" objects with a very weak bar are noted $SABO$. The symbol SO , used for SAO in HUBBLE's notation, is now used for a lenticular object which cannot be more precisely classified as either SAO or SBO ; this is often the case for edgewise objects¹.

Two main *varieties* are recognized in each of the lenticular and spiral families, the "annular" or "ringed" type, denoted (r), and the "spiral" or "S-shaped" type, denoted (s). Intermediate types are noted (rs). In the "ringed" variety the structure includes circular (sometimes elliptical) arcs or rings (SO) or consists of spiral arms or branches emerging tangentially from an inner circular ring (S). In the "spiral" variety two main arms start at right angles from a globular or little elongated nucleus (SA) or from an axial bar (SB).

The distinction between the two families A and B and between the two varieties (r) and (s) is most clearly marked at the transition stage SO/a between the SO and S classes. It vanishes at the transition stage between E and SO on the one hand, and at the transition stage between S and I on the other (cf. Fig. 3).

Four *sub-divisions* or *stages* are distinguished along each of the four spiral sequences $SA(r)$, $SA(s)$, $SB(r)$, $SB(s)$, viz. "early", "intermediate" and "late" denoted a , b , c as in the standard classification, with the addition of a "very late" stage, denoted d . Intermediate stages are noted Sab , Sbc , Scd . The transition stage towards the magellanic irregulars (whether barred or not) is noted Sm , e.g. the Large Magellanic Cloud is $SB(s)m$.

Along each of the non-spiral sequences the signs $+$ and $-$ are used to denote "early" and "late" subdivisions; thus E^+ denotes a "late" E , the first stage of the transition towards the SO class². In both the SAO and SBO sub-classes three stages, noted SO^- , SO^0 , SO^+ are thus distinguished; the transition stage between SO and SA , noted SO/a by HUBBLE, may also be noted SA^- . Notations such as SA^+ , Sb^- , etc. may be used occasionally in the spiral sequences, but the distinction is so slight between, say, SA^+ and Sb^- , that for statistical purposes it is convenient to group them together as Sab , etc. Experience shows that this makes the transition subdivisions, Sab , Sbc , etc. as wide as the main subdivisions, SA , SB , etc.³.

¹ The symbol (S) may be used when even the distinction between S and SO is no longer possible, as in the case of very small, poorly resolved objects.

² An example is NGC 3115 which, although selected as the type object for $E7$ in HUBBLE's original scheme, shows an absorption marking and a beginning of nuclear differentiation on 200 inch plates.

³ No E^- , nor I^- or I^+ are at present clearly recognized, but such notations may later prove useful for a still finer classification based on accurate surface photometry.

The classification of “irregulars” is still somewhat in doubt; objects obviously related to the magellanic type, but which do not show clearly the characteristic spiral structure, are noted $I(m)$; those with an elongated core and asymmetrical branches are probably later stages of the SB sequences, e.g. NGC 4449, 6822, etc., while those more nearly symmetrical and without bar-like core are probably later stages of the SA sequences, e.g. IC 1613, IC 2574, etc. (see Plate VII) the distinction vanishes in the ultimate, chaotic dwarf stages of low surface brightness, e.g. the Wolf-Lundmark nebula or the Sextans system¹.

A faint, *outer* ring-like structure is often observed in lenticulars and early-type spirals which is not clearly related to a specific type, i.e. with some variants it appears about equally in all four sequences near the transition stage SO/a . This particularity which seems therefore more characteristic of a certain stage of evolution than of any definite line of evolution, is denoted by an (R) preceding the symbol of the class. Examples of $(R)SA$ are NGC 1068 (M 77), NGC 4736 (M 94), NGC 7217, etc. of $(R)SB$: NGC 1291, 1326, 2859, etc. Some objects have both an outer (R) structure and an inner (r) pattern; a good example is NGC 6753, noted $(R)SA(r)ab$ (Plate V, 22).

In the spiral sequences supplementary data of some interest are the *multiplicity* of the spiral pattern and the *character*, “massive” or “filamentary” of the spiral arms. These are included whenever possible as subscripts respectively after the family and variety symbols. Thus M 33 is noted $SA(s)_2c_m$ and described as “an ordinary, late-type, S-shaped spiral with two main massive arms”. When one or more additional and weaker arms are present this is noted $2+1$ (e.g. M 99), $2+2$ (e.g. M 51), etc. An asymmetrical, regular spiral with one arm stronger than the other as is often the case among late-type barred spirals (e.g. NGC 7479) is noted $1+1$ (not 2). Branching arms not clearly assignable to a definite multiplicity are noted 1^+ , 2^+ , etc. as the case may be; this is a frequent occurrence among late-type spirals. A complicated spiral pattern of multiplicity higher than 4, as often observed in late-type spirals, is simply noted n (e.g. NGC 2903, 7793); this, however, occurs also among early-type spirals of the ringed variety (e.g. NGC 4736, 7217).

Edgewise systems which can usually be classified only as $E7$, SO , $S(a, b, c, d)$ or I are noted (sp) for “spindle”. For ellipticals the traditional notation $E0$, $E1, \dots E7$ may be preserved, although it is not homogeneous with the rest of the classification scheme and notation system used for the other classes; it is really almost superfluous when measured dimensions and flattening are given.

An advantage of the present notation system is that it is possible to retain significant information on objects poorly resolved by dropping one or more symbols which from right to left refer to stages, varieties, families and classes of increasing generality. Thus S may be either SA or SB , SO either SAO or SBO , SA is either $SA(r)$ or $SA(s), \dots$, Sa is either SAa or $SBa, \dots$, etc. Hence the nature and number of symbols used can be selected according to the amount of detail visible in any given object with any particular telescope. Conversely the degree of completeness in the symbolism gives an indication of the relative degree of resolution available; this, of course, depends on both the size of the telescope and the size of the nebula.

¹ Objects of low surface brightness (when it is not obviously due to local obscuration, e.g. as for IC 10, IC 342 etc.) are almost invariably dwarf systems; this particularity may be noted by a (d) preceding the symbol of the class which is usually either E or I . The existence of dwarf lenticulars and spirals remains to be demonstrated. Criteria are still lacking for the assignment of objects to the giant and supergiant groups on purely morphological grounds.

b) Pairs and multiplets.

These are discussed in detail by ZWICKY. It is nevertheless of some importance for the present discussion to determine the separation between neighboring objects within which their mutual interaction begins to cause significant structural distortion and thus sets a limit to the validity of the classification system established in the first instance for isolated galaxies. A rough classification of pairs is as follows:

$P(a)$: pairs of widely separated objects showing no clear sign of interaction (occasional optical pairs are included in this group);

$P(b)$: close pairs showing clear signs of interaction, such as tidal distortion of their outlying parts;

$P(c)$: colliding pairs where the "main bodies" are in actual contact and show extensive disruption of their internal structure.

Many parameters are obviously involved, such as separation, diameters, masses, velocities, etc. in the interaction of two galaxies. Only the first two are more or less directly accessible in most cases and more precisely the ratio x' of the projected separations $s'_{1,2}$ between the nuclei G_1, G_2 of a pair to the sum of their apparent photographic radii r_1, r_2 . The true separation s is not known in individual cases, but for a random distribution of orientations of $\overline{G_1 G_2}$ with respect to the line of sight the mean value of s'/s is $\pi/4 \approx 0.79$. For 40 pairs south of -35° observed with the Reynolds reflector on Mt. Stromlo¹ the following mean values were obtained:

$$P(a) \quad \bar{x}' = 3.65 \quad (\text{range: } 1.6 - 9:) \quad n = 30,$$

$$P(b) \quad \bar{x}' = 1.30 \quad (\text{range: } 0.6 - 1.6) \quad n = 8,$$

$$P(c) \quad \bar{x}' = 0.3: \quad (\text{range: } 0.2: - 0.4:) \quad n = 2.$$

Allowing for scale factors the average projected and actual separations are:

$$P(a) \quad \bar{s}' = 18 \text{ kpc} \quad (\text{range } 8 \text{ to } 45 \text{ kpc}) \quad \bar{s} = 23 \text{ kpc},$$

$$P(b) \quad \bar{s}' = 6.5 \text{ kpc} \quad (\text{range } 3 \text{ to } 8 \text{ kpc}) \quad \bar{s} = 8 \text{ kpc}$$

on a distance scale in which HUBBLE's expansion parameter is 180 km/sec per Megaparsec. The mean critical separation ($x' \approx 1.6$, $x \approx 2.0$) within which tidal distortion becomes notable is then of the order of 10 kpc. Among the bright galaxies south of -35° only 10 out of more than 400, or less than 2.5%, are within this limit. Hence, the revised classification should be applicable to at least 97 or 98% of the external galaxies, of which about 95% can be included in the normal scheme of Fig. 2 and the remaining 2 or 3% must still be regarded as special or "peculiar" cases reserved for further analysis.

c) Frequency of revised types.

A classification of over 200 southern galaxies in the revised system [80] yields the following apparent relative frequencies for a sample essentially selected according to apparent magnitude and surface brightness (all objects listed in the Shapley-Ames Catalogue south of -35°).

¹ The photographic diameters used in the Mt. Stromlo survey [80] are corrected for tilt effects, i.e. reduced to "face-on", and define a system of dimensions intermediate between those of REINMUTH [62] and of SHAPLEY [66].

The regular spirals SA , SB make up just over fifty percent of the sample, ellipticals and lenticulars close to a quarter each. The discrepancies between earlier statistics (cf. Sect. 1b) may therefore have hinged mainly on whether

Table 1. *Frequency of revised types.*

Class/Family	E	SO	SA	SB	I	Pec
Frequency (%) . .	23.4	21.0	24.4	26.3	3.4	1.5

the lenticulars were counted as ellipticals or spirals the ratio S/E changing correspondingly from 1/1 to 3/1. Note also that the ordinary spirals appear no longer more abundant than the barred spirals despite the inclusion of intermediate types (SAB) with the SA group in the table.

Table 2. *Frequency of sub-divisions.*

Type	E	E/SO	SO	SO/a	Sa	Sab	Sb	Sbc	Sc	Scd	Sd	Sm
Frequency	22.0	9.5	10.5	9.0	4.5	6.5	7.5	7.5	10.5	8.5	2.0	2.0

The detailed frequency in each of the sub-divisions along the regular classification sequence is given in Table 2, in which $E/SO = E^+$, SO^- and $SO/a = SO^+$, Sa^- .

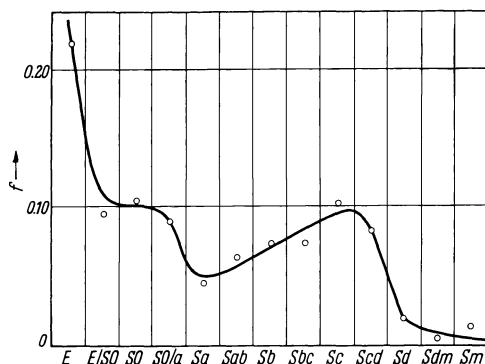


Fig. 4. Frequency distribution of revised types in the Mount Stromlo survey.

A plot of the frequencies (Fig. 4) gives a fairly smooth curve indicating maxima at Sc and E separated by a minimum at Sa . The low apparent frequency of the Sd , Sm stages is certainly due to a large extent to the low absolute luminosity and surface brightness of galaxies at these stages, but it is not yet possible to correct the apparent frequencies for such selection effects. On the other hand the high frequency of ellipticals cannot be due in general to an abnormally high absolute luminosity, so that it probably reflects the actual great abundance of this class¹.

¹ It should be emphasized that these provisional data based on a rather small sample of 200 objects are still subject to further revision; more definite and detailed results will eventually be forthcoming through the current reclassification of 1250 bright galaxies in the Shapley-Ames Catalogue.

II. Morphology.

a) Qualitative morphology.

4. Description of typical examples. The following eleven plates illustrate the revised classification system and will serve to support the description of the main morphological characteristics used as classification criteria. Since the revised three-dimensional classification recognizes at least 16 stages along the sequence from *E* to *Im* and 9 or more pure or mixed types at each stage across it (at least in the spiral sequences), a fully illustrated description, allowing also for tilt and orientation effects, would require several hundred examples. However, the main criteria used to define the position of any normal galaxy along and across the classification volume can be illustrated with only a fraction of the total number of possible cases.

Plates I and II describe the main criteria used to distinguish between families and varieties in a central cross-section near the stage *Sb*.

Plates III and IV illustrate the ellipticals and principal stages of the lenticular sequences.

Plates V and VI illustrate the four main stages along each of the four principal sequences of ordinary and barred spirals.

Plate VII shows the (magellanic) irregular later stages of the spiral sequences.

Plates VIII and IX illustrate the appearance of edgewise system of the lenticular, spiral and irregular classes.

Plate X illustrates the effects of tilt and orientation on the appearance of *SB(s)a* or *SB(r)a* spirals.

Plate XI shows some peculiar lenticulars and (non-magellanic) irregulars of the M 82 type.

In order to minimize as far as possible the complications arising from differences in scale and resolution, several sources of illustrations were used as follows:

1. Mt. Palomar 200-inch reflector (P 200'') and 48-inch telescope (P 48'').
2. Mt. Wilson 100-inch and 60-inch reflectors (W 100''; W 60'')¹.
3. Mt. Stromlo 74-inch and 30-inch reflectors (S 74''; S 30'').
4. Isaac Roberts 20-inch reflector (IR 20'')².

For technical reasons all photographs are reproduced here as inverted negative prints, i.e. are mirror images of the objects as they appear on the sky, and the orientation with respect to the celestial coordinates is arbitrary.

The classification criteria are described in relation to the three main parts of a galaxy defined by HUBBLE:

a) the *nucleus*, i.e. the very small, very bright central condensation often sharply defined as the centre of symmetry of the structure; it is round in *SA*, elliptical in *SB*.

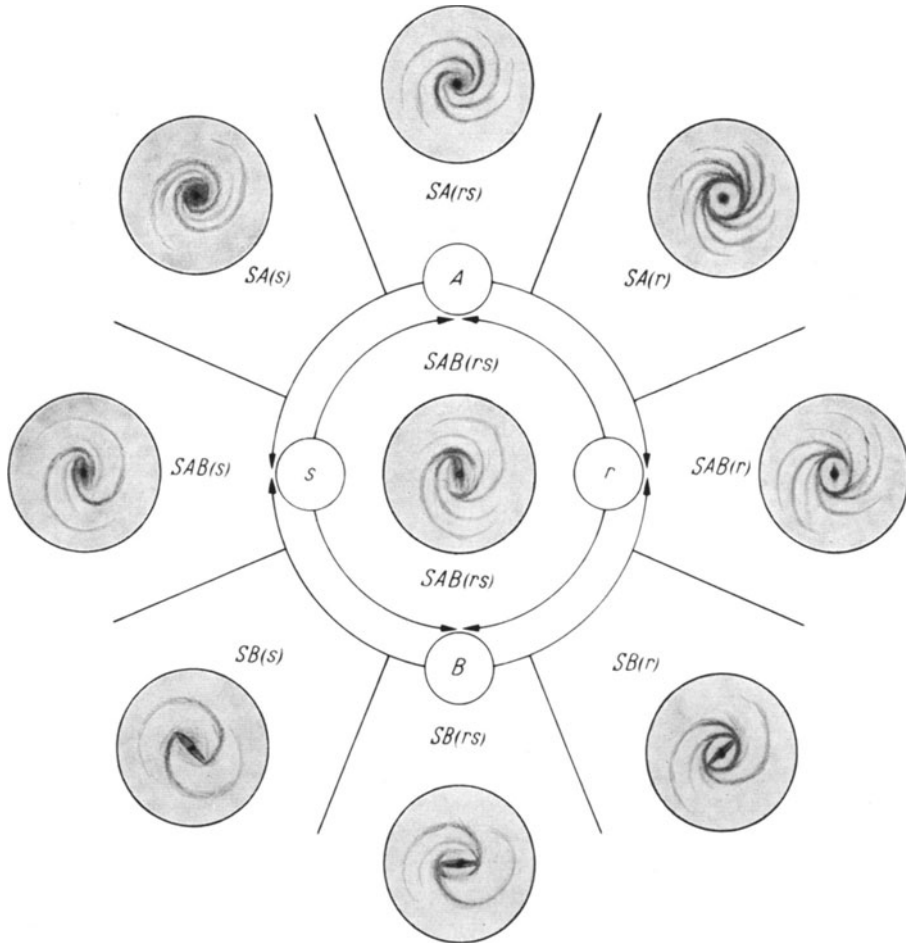
b) the *lens*, smooth, bright and sharply defined in the lenticulars and early spirals, being crossed by the bar in *SB* and brighter at the edge in *S(r)*.

c) the *envelope*, generally faint and smooth with indefinite outer boundary in the lenticulars, brighter and occupied by the spiral arms in spirals. The outer (*R*) structure appears often in its outer parts near the stage *SO/a*.

The distinction between these three regions is most definite near the middle of the classification sequence (*SO/a*) and vanishes at *E* and *Im*.

¹ The Mt. Wilson and Palomar photographs are reproductions from the "Hubble Memorial Volume" generously communicated in advance of publication by Dr. A. R. SANDAGE.

² The Isaac Roberts photographs are from direct prints of the original plates, now stored at the Paris observatory and kindly made available some years ago by Prof. P. COUDERC.



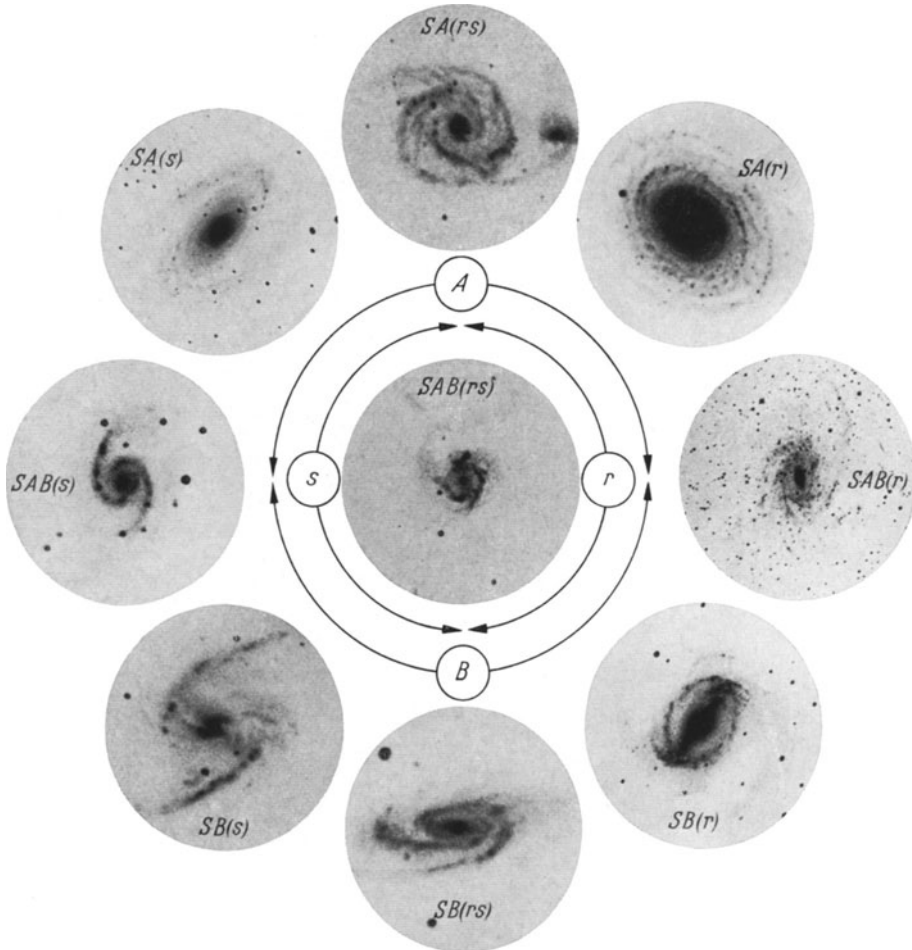
Plates I and II. Cross-section of classification volume near stage Sb.

The spirals of the ordinary family *SA* are above, of the barred family *SB* below; the ringed variety *S(r)* is to the right, the S-shaped variety *S(s)* to the left. The mixed types *SAB(rs)* are in the centre. There is a continuous transition between each type and only the main forms are illustrated; actual objects can occupy any intermediate position in the plane (volume).

The main classification criteria are as follows:

SA(r): has a small, sharp, very bright and round nucleus isolated in the centre of a circular ring at the edge of which emerge many tightly wound filamentary spiral arms or arcs; the nucleus and ring often merge and disappear in the over-exposed image of the central bulge (as illustrated), but the high multiplicity of the spiral pattern is usually sufficient for identification; a weak outer ring (*R*) made up of many closely coiled spiral arcs is often present in early stages of this sequence. Examples are NGC 488 (illustrated from P 200'), 7217, 6753 (Plate V, 22); other examples: NGC 4736 (M 94) (Plate V, 21), *Sab* and 5055 (M 63), *Sbc*.

SA(rs): has a fairly small, fairly sharp, bright and round nucleus in the centre of a diffuse lens or bulge out of which two main arms and two or more additional, weaker arms emerge tangentially; the two main arms simulate an incomplete ring around the lens; weak outer arms simulate an (*R*) structure in the early stages. Examples are NGC 1068 (M 77), 3147, 4237, possibly NGC 3521, 4800, 7079, 7590. NGC 5194 (M 51) (illustrated from IR 20'') is not a very good example as it is more nearly *Sbc* and intermediate between *SA(rs)* and *SA(s)*.



SA(s): This is the typical, regular logarithmic spiral; it has a fairly large, diffuse, round nucleus extending smoothly into the circular or little elongated lens which shows some spiral pattern of dark matter; two main spiral arms with occasional branching or weaker secondary arms start at the rim of the lens. Examples are NGC 3031 (M 81), *Sab* (illustrated from IR 20''), 4569, 7205 (Plate V, 18), 7331. One diameter of the lens is occasionally slightly brighter and longer and the outer arms may have a tendency to "return" to the lens, indicating a transition toward the *SAB(s)* type; examples are NGC 224 (M 31), *Sb* and 4321 (M 100), *Sbc*.

SAB(s): Has a fairly small, bright, elongated nucleus crossed by a weak, twisted dark lane in the centre of a weak and broad bar marking the major axis of a faint, diffuse and elongated lens with much spiral pattern of dark matter having a tendency to run parallel to the bar. Two main arms emerge near the extremities of the bar along a smooth, curved path turning sharply just outside the lens; their faint extremities tend to return to the lens along an almost circular outer loop, simulating an (*R*) structure in the early stages. Examples are NGC 1566 (illustrated from S 30''), NGC 5236 (M 83), *Sbc*, 4579, *Sab* and 7392.

SB(s): This is the typical "S-shaped" barred spiral; it has a small, very bright and elongated nucleus distorted by a strong, twisted dark lane as it crosses over from one side to the other of the strong, narrow bar marking the major axis of a much elongated lens. Two strong, main arms start sharply at right angles to the bar at both ends and return faintly to it after completing a turn of a quasi-circular loop. Examples are NGC 1097,

1300, 1365 (illustrated from S 30''), 5383. Later stages in this sequence are markedly asymmetrical as NGC 7479, *Sc* and 7741, *Scd* (Plate VI, 31). Some, like NGC 1300, have a third, fainter arm forming a half-ellipse close to the lens and indicating a transition toward the *SB(rs)* type.

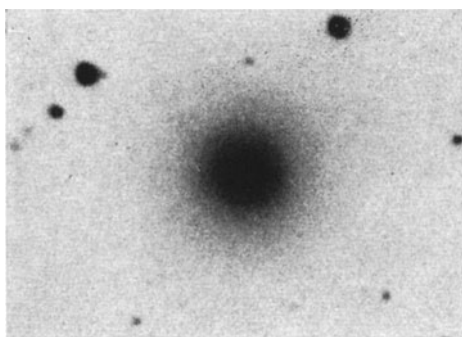
SB(rs): has a fairly small, very bright and elongated nucleus crossed by a twisted dark lane, as in *SB(s)*, in a very strong, very narrow bar marking the major axis of an elongated lens whose rim is brighter, especially near the extremities of the bar; this produces a characteristic "dash-dot in brackets" pattern, thus (— o —), from which emerge two main spiral arms with faint additional branches near the rim of the lens. Examples are NGC 4548, 4593, 7124 (illustrated from S 74''). Some, like NGC 4349, *Sab*, having a brighter lens and more continuous rim mark the transition toward *SB(r)*.

SB(r): This is the typical " Φ -shaped" barred spiral. It has a fairly large, elongated nucleus with weak spiral dark lane in a strong, narrow bar along the major axis of an elliptical ring marking the edge of the lens. Two main arms start tangentially from the ring near the extremities of the bar (i.e. at right angles to the bar); one or two fainter arms branch out from breaks in the ring near its minor axis. In early stages the main arms tend to form an outer, circular (*R*) structure, as in NGC 1433 (Plate VI, 25); in later types the breaks in the ring tend to produce slightly hexagonal shapes. Examples are NGC 1433 (illustrated from S 74''), *Sa* or *Sab*, 3185, 3351, *Sab* (Plate VI, 26), 2523, *Sbc* (Plate VI, 27).

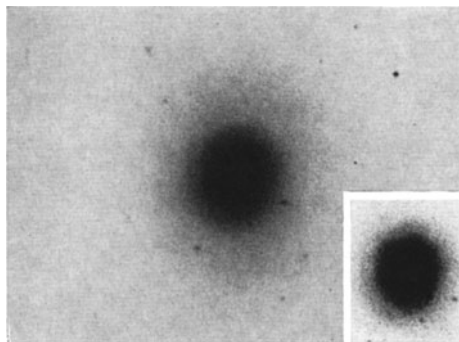
SAB(r): has a fairly small, little elongated nucleus in a fairly broad and faint bar marking the major axis of a little elongated ring from which several spiral arms branch out. The main arms have a slight tendency to "return" inwards. Examples are NGC 1832, 7531, *Sab*, 6744, *Sbc* (illustrated from S 30''). Some, like NGC 6902, 6935, *Sa*, 6937, have only very faint traces of a bar and mark the transition towards *SA(r)*.

SAB(rs): this is the most general mixed type involving all possible transitions between the main typical patterns. An almost infinite variety is possible here, but for classification purposes the main characteristics of this hybrid type are: a small bright nucleus in a broad, diffuse bar with some spiral structure in the lens. The bar crosses a nearly circular or often hexagonal pseudo-ring formed by the inner sections of the spiral arms. A very good example is NGC 4303 (M 61), *Sbc* (illustrated from IR 20''); other examples are NGC 3145, 6814. At a later stage this mixed structure is well illustrated by the central regions of NGC 5457 (M 101) and NGC 6946, both *Scd*.

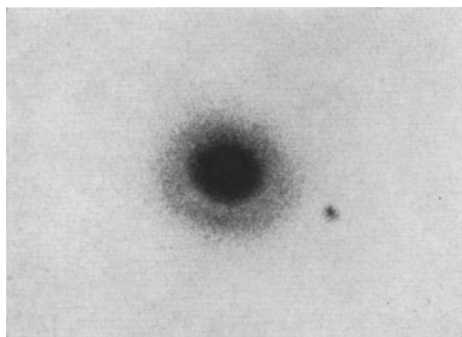
Plate III. Ellipticals and ordinary lenticulars.



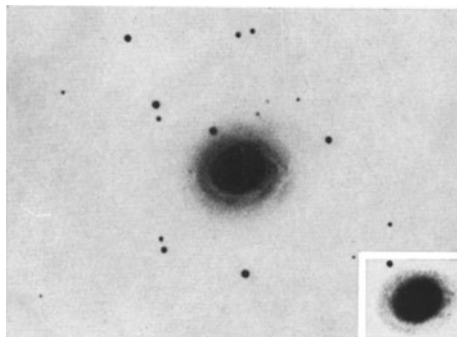
1. NGC 7144 (S 74''): *EO*. The appearance of a distinct nucleus is due to a photographic effect; there is actually a smooth, continuous decrease of luminosity from the centre outwards.



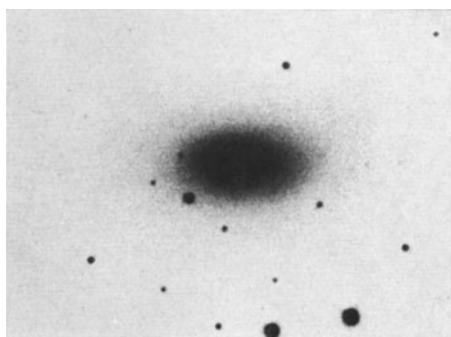
2. NGC 5273 (P 200''): *SA(s)O⁻*. Traces of spiral structure in the smooth lens and envelope surrounding the large nucleus; little or no dark matter. With lower resolution lens and envelope would simulate a late elliptical; compare with 1 and 5.



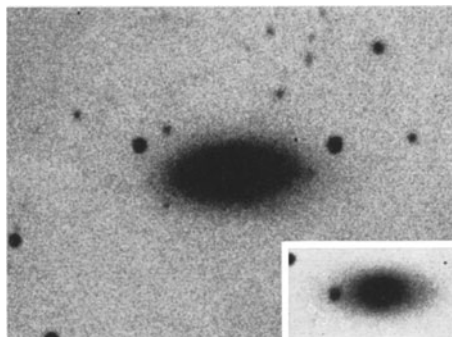
3. NGC 4459 (W 100''): $SA(s)O^0$. A weak ring or whorl of dark matter appears in the lens close to the large nucleus. The envelope does not show up on this view taken with a short exposure. Compare with 4 and 7.



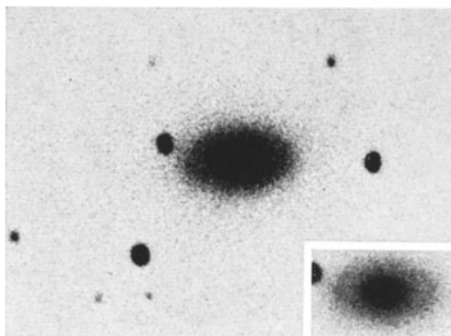
4. NGC 2855 (P 200''): $SA(s)O^+$. The ring of dark matter outlines a weak smooth spiral structure around the large and bright nucleus. Compare with 3 and 8.



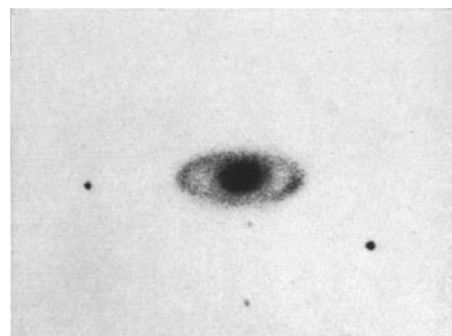
5. NGC 7457 (W 100''): $E^+ 5$. The merest traces of differentiation between nucleus and lens appear at the first transition stage between E and SO . Compare with 1, 2 and 6.



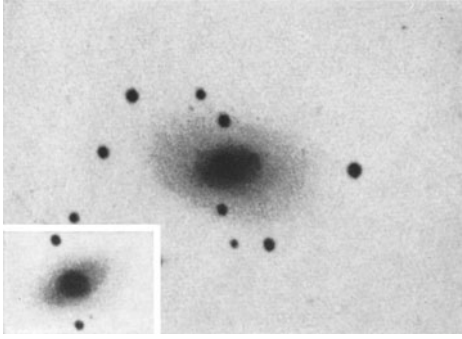
6. NGC 7166 (S 74''): $SA(r)O^-$. The lens and envelope still simulate an elliptical, but a small nucleus appears in the lens. Note size of nucleus; compare with 2 and 5.



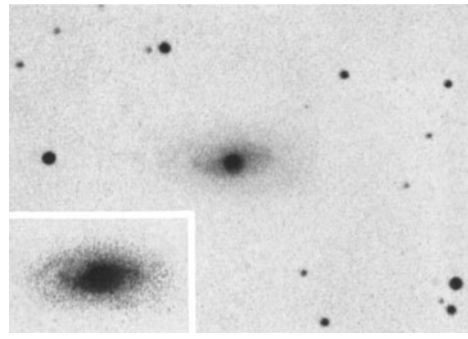
7. NGC 1553 (S 30''): $SA(r)O^0$. A weak trace of ring appears at the edge of the lens. Note small, distinct nucleus in lens and faint, outer envelope. Compare with 3, 6 and 8.



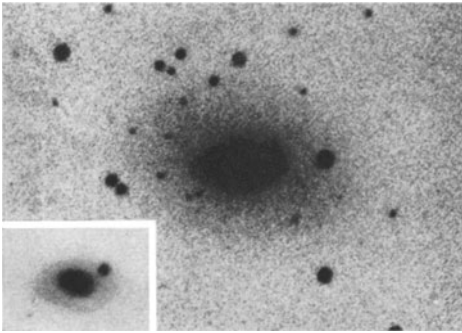
8. NGC 7702 (S 74''): $SA(r)O^+$. A definite ring, well separated from the small nucleus appears at the edge of the lens. The very faint envelope does not show up on this image of short exposure. Note asymmetry of nucleus and ring near minor axis indicating the presence of dark matter in the equatorial plane. Compare with 4 and 7.



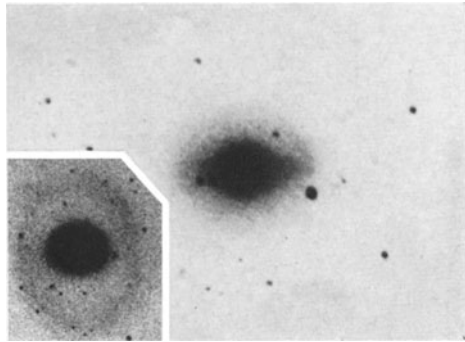
9. NGC 7079 (S 74''): $SB(s)O^0$. Small bright nucleus in short and bright bar surrounded by very faint traces of spiral arcs in lens; extensive envelope. Compare with 12 and 13.



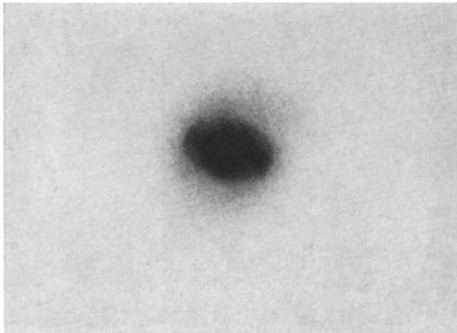
10. NGC 1617 (S 30''): $SB(s)O^+$. Very small nucleus in short and fainter bar curving into spiral arms or arcs in lens and envelope. The spiral pattern is blurred by insufficient resolution here. Compare with 9 and 14.



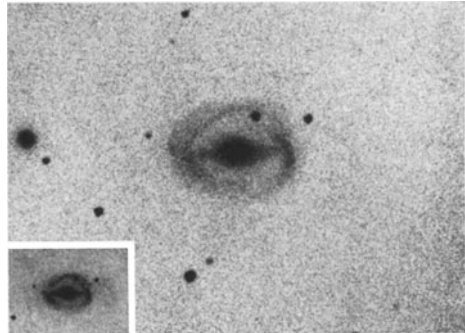
11. NGC 6873 (S 74''): $SAB(rs)O^0$. Fairly small nucleus in elongated lens with dark ring or crescent, surrounded by extensive, bright envelope. Compare with 3, 7 and 9.



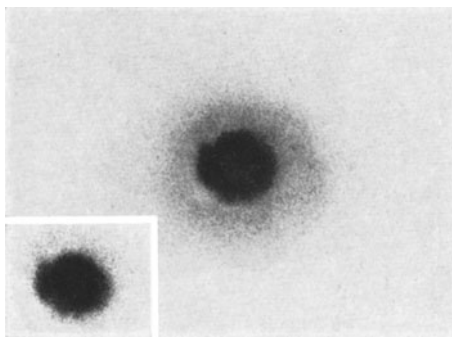
12. NGC 1291 (S 30''): $(R)SB(s)O^+$. Fairly large nucleus on weak bar marking major axis of little elongated bright lens. Faint outer ring structure from which emerge two very weak spiral arms or arcs. No ring structure in lens. Compare with 9, 10 and 16.



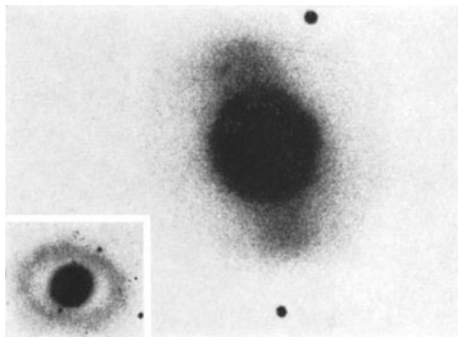
13. NGC 4262 (P 200''): $SB(r)O^0$. Small nucleus in short, stubby bar (stronger near extremities) in bright diffuse lens. The nucleus appears large on this image taken in yellow light. Compare with 9, 15 and 16.



14. NGC 1512 (S 30''): $SB(r)O^+$. Small, elongated nucleus on narrow, bright bar marking the major axis of elliptical ring; the ring is stronger near the extremities of the major axis. Traces of spiral arms or arcs in weak envelope outside ring. Compare with 8 and 25; NGC 1510 (EO) is at left near edge of field.



15. NGC 3032 (P 200''): $SAB(r)O^{\circ}$. Fairly large, elongated nucleus on short, bright stubby bar, surrounded by dark ring or arcs in bright lens. The envelope does not show up on this print. Compare with 3, 7 and 13.

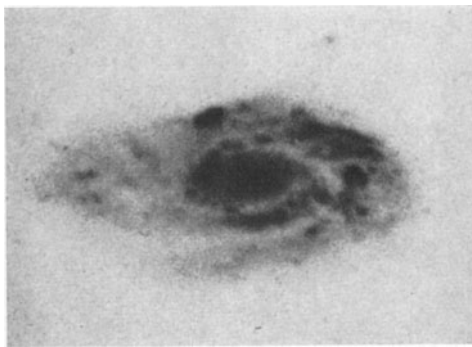


16. NGC 2859 (P 200''); inset P 48'': $(R)SB(r)O^{\circ}$. Fairly large nucleus on weak bar marking major axis of little elongated lens, slightly stronger near rim where bar appears brighter. Faint outer ring structure from which emerge two very weak spiral arms or arcs. No spiral structure in lens. Compare with 12, 14 and 15.

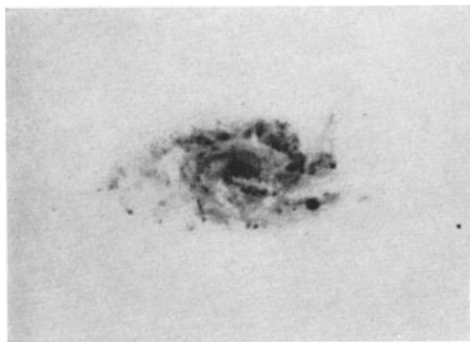
Plate V. The two main sequences of ordinary spirals.



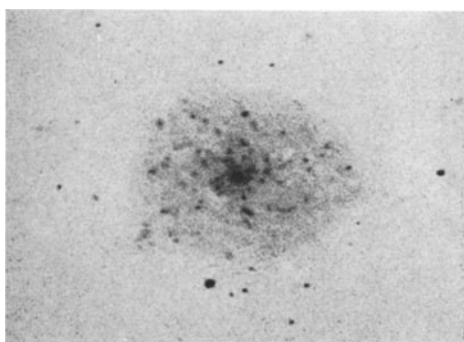
17. NGC 4569 = M 90 (P 200''): $SA(s)ab$. A large central bulge from which emerge two smooth and massive regular arms which are closely wound and separated by dark lanes. Faint outer spiral extensions showing tendency to return towards central lens. Compare with 4, 21 and 39.



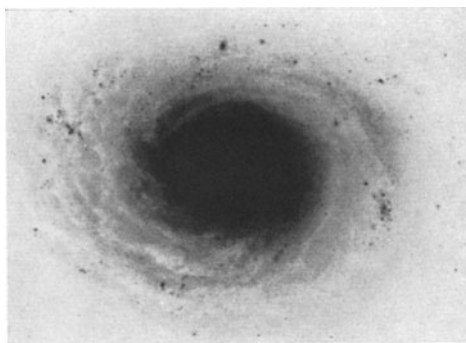
18. NGC 7205 (S 74''): $SA(s)b.c$. Smaller central bulge with inner nucleus from which emerge two main massive and knotty regular arms; these are less closely wound and show some branching. Compare with 17, 18 and 40 or 41.



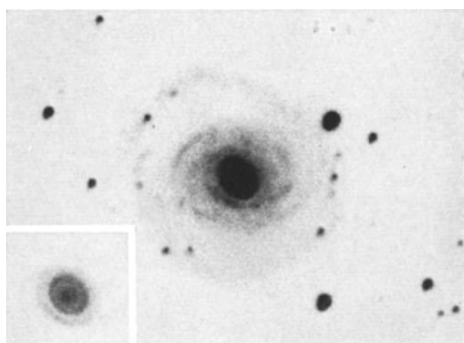
19. NGC 1084 (W 100''): $SA(s)c$. A small central nucleus from which start two main arms with much irregular structure, bright knots and emission objects. Only the brighter, inner regions are shown in this short-exposure image. Compare with 18, 20, 23 and 42.



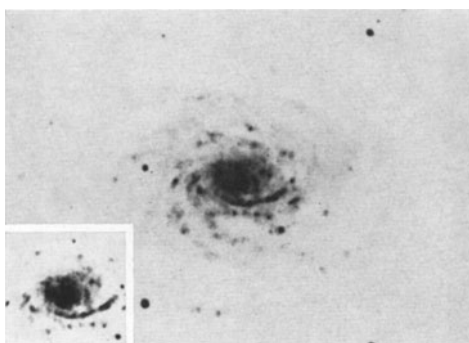
20. NGC 7793 (S 30''): $SA(s)d$. A very small central nucleus, resembling a globular cluster, is surrounded by several irregular and broken spiral arms with very many knots. Note that the arms still start from the nucleus. The mean surface brightness decreases steadily from Sa to Sd . Compare with 19, 24 and 33.



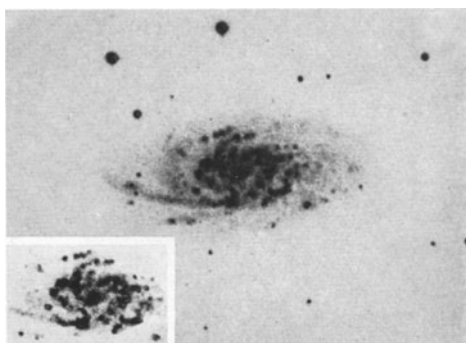
21. NGC 4736 = M 94 (P 200''): (*R*) *SA* (*r*) *ab*. Only the complex spiral pattern of high multiplicity in the lens appears in this reproduction. The small nucleus and inner ring (*r*) surrounding it are lost in the overexposed central bulge; the outer, smooth ring structure (*R*) extending beyond the limits of the figure is not shown either. Compare with 8, 17 and 22.



22. NGC 6753 (S 74''): (*R*) *SA* (*r*) *b*. The inset shows the nucleus and bright inner ring (*r*) from which emerge the complex spiral pattern in the lens, only partly resolved here; the outer ring (*R*), which begins to break up into spiral arcs with bright knots is faintly connected to the inner spiral pattern in two diametrically opposite points. Compare with 8, 21 and 23 (allowing for differences in scale and resolution).

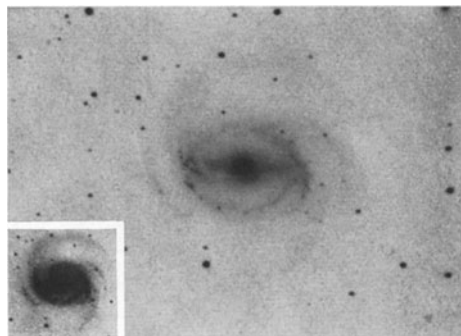


23. NGC 5962 (W 100''): *SA* (*r*) *c*. The inset shows the nucleus and inner ring which begins to break up at this stage and from which extend the many knotty and filamentary arms; only weak traces remain of the outer ring structure. Compare with 19, 22 and 24.

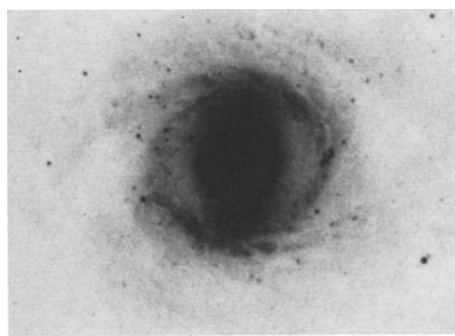


24. NGC 6643 (W 60''): *SA* (*r*) *d*. The inset shows the small nucleus separated from broken remnants of the inner ring extending into the complex, irregular spiral pattern. Compare with 20, 23 and 33.

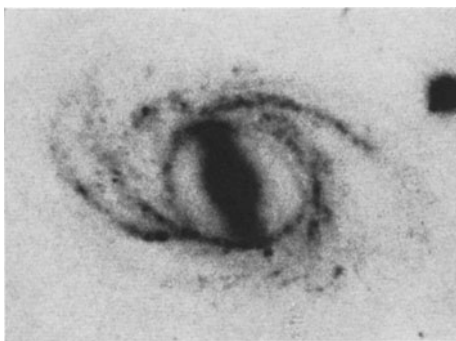
Plate VI. The two main sequences of barred spirals.



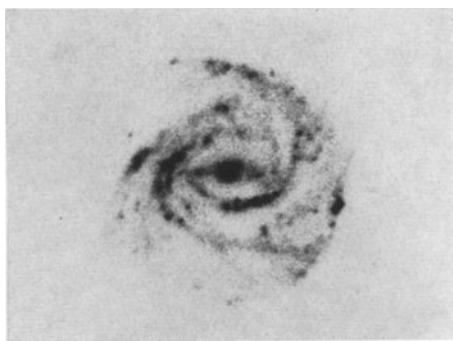
25. NGC 1433 (S 74''): (*R'*) *SB* (*r*) *a*. A small, elongated nucleus whose major axis is tilted at 45° to a long, narrow bar along which run filaments of dark matter. The bar marks the major axis of an elliptical ring from which emerge two faint main arms near the extremities of the bar and two weaker branches forming an incomplete outer ring (inset). Compare with 14, 26 and 29.



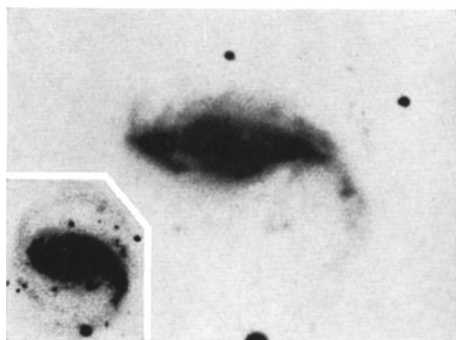
26. NGC 3351 = M 95 (P 200''): (*R'*) *SB* (*r*) *ab*. The nucleus and bar are enhanced on this photograph taken in yellow light; the elliptical ring appears circular or slightly hexagonal by projection foreshortening; note the two main arms emerging from ring near the extremities of the bar and weaker branches. Compare with 25, 27 and 30.



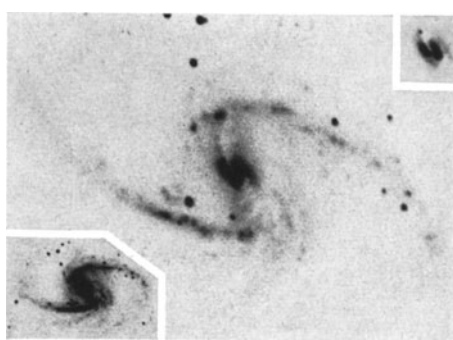
27. NGC 2523 (P 200''): $SB(r)bc$. The nucleus and bar are still strong, but the ring is narrower and knotty; two main spiral arms, narrow and knotty (one with branching) emerge from the ring near the extremities of the bar; additional, weaker arms start near minor axis of elliptical ring (here appearing circular in the projection). Note slight asymmetry. Compare with 26, 28 and 31.



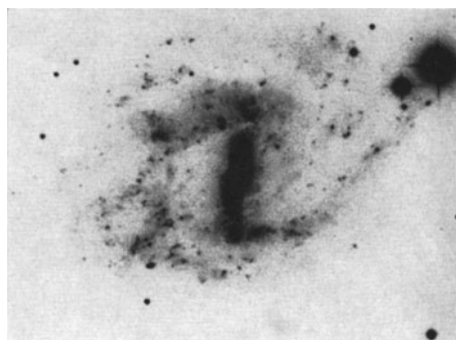
28. NGC 3367 (P 200''): $SB(r)cd$. The nucleus and bar are smaller and fainter; the elliptical ring is weaker and begins to break up into knots; the arms are branching and filamentary; note strong asymmetry. Compare with 24, 27 and 31.



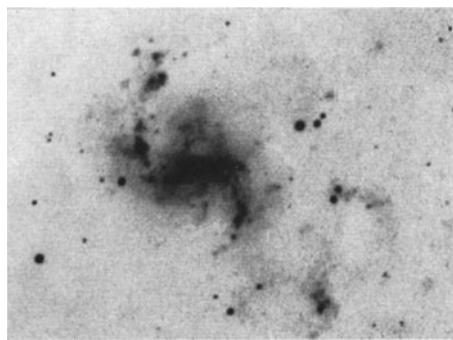
29. NGC 7552 (S 74''): $(R')SB(s)a$. A small, elliptical nucleus in a bar-like, elongated lens with much internal structure marked by dark lanes; two main spiral arms (one stronger) emerge from the extremities of the bar and return each to the opposite extremity after a halfturn, thus simulating an outer ring structure. Compare with 10, 23, 30 and Plate X.



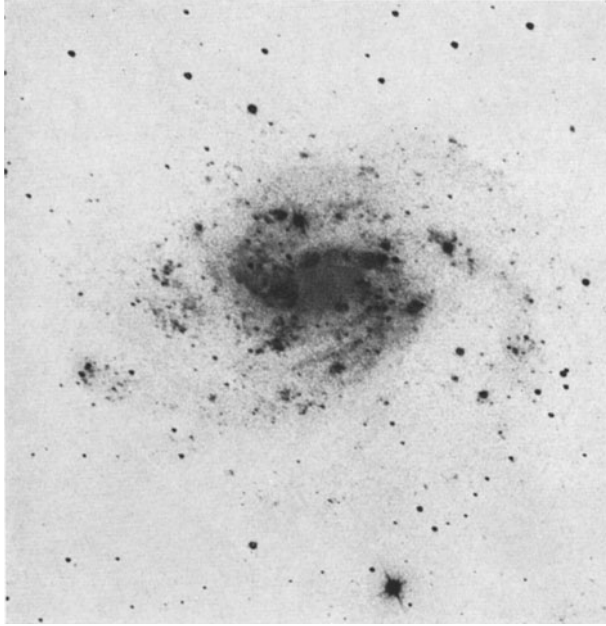
30. NGC 1365 (S 30''): $(R')SB(s)b$. The upper right inset shows the complex structure of the nucleus due to the "crossing-over" of the main dark lane running along the bar which is seen here foreshortened in the projection; the two bright and knotty main spiral arms emerge at right angles to the bar and return to it after a half-turn (lower inset), thus simulating an outer ring. Compare with 27, 29, 31 and Plate X.



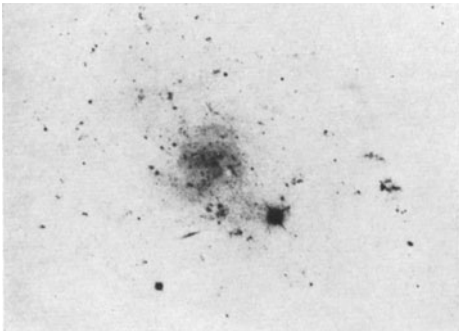
31. NGC 7741 (P 200''): $SB(s)cd$. The nucleus becomes inconspicuous in the bar (foreshortened in the projection); the two main, heavy arms show much branching and clumpiness; one is much stronger than the other, producing a characteristic asymmetry. Compare with 27, 30 and 32.



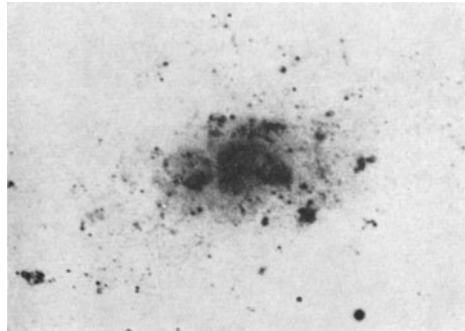
32. NGC 1313 (S 74''): $SB(s)d$. Only a very small nucleus remains in the center of the bar from which emerge two short arms broken up into clumpy sections. The rest of the spiral pattern is weak, very irregular and strongly asymmetrical; note also asymmetrical halfring on one side of bar. This is the transition stage towards the magellanic irregulars. Compare with 31, 36 and 37.

Plate VII. The irregular later stages of the spiral sequences.

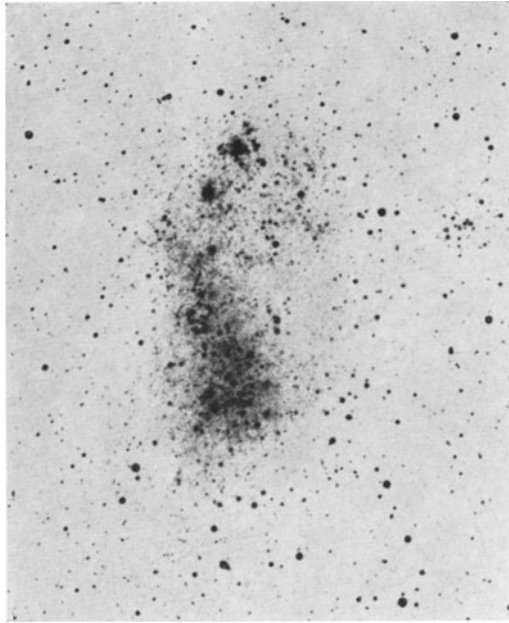
33. NGC 300 (S 30''): *SA(s)d*. A very small, round nucleus, similar to a globular cluster, lies in the middle of the central region from which emerge several, highly resolved and branching arms of low surface brightness. Compare with 20 and 34 NGC 300 is very similar to M 33 but of slightly later type.



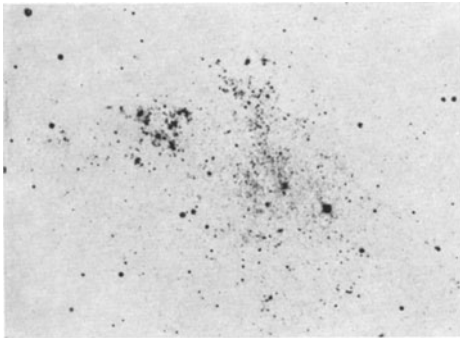
34. NGC 45 (P 200''): *SA(s)dm*. Only the central regions of this object of very low surface brightness are visible on this print; the nucleus is very small and faint in a central, amorphous mass from which emerge very faint, irregular and branching arms. Compare with 33 and 35.



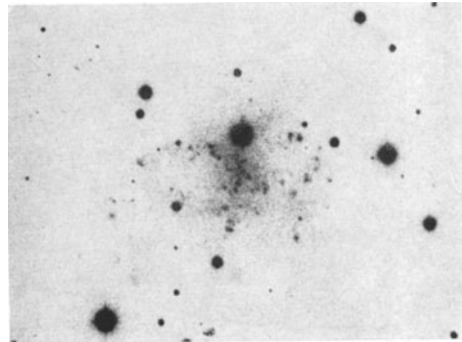
35. NGC 5204 (P 200''): *SA(s)m*. Only a weak spiral pattern remains in the highly resolved, irregular, but symmetrical distribution of supergiant stars and emission knots surrounding the central condensation. Compare with 34 and 37.



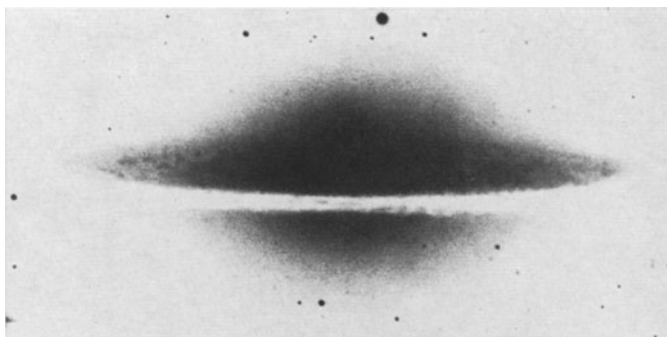
36. Small Magellanic Cloud (AE 2''): $SB(s)m_p$. The spiral pattern is reduced to the axial bar from which emerge one bright, knotty and asymmetrical arm (above, right)—including many supergiants—and one faint, smooth arm (lower left, invisible on print) including no bright supergiant. Traces of embryonic arms appear at the other extremities of the bar. The peculiarity is the asymmetrical distortion towards the Large Cloud (to right). Compare with 32 and 37.



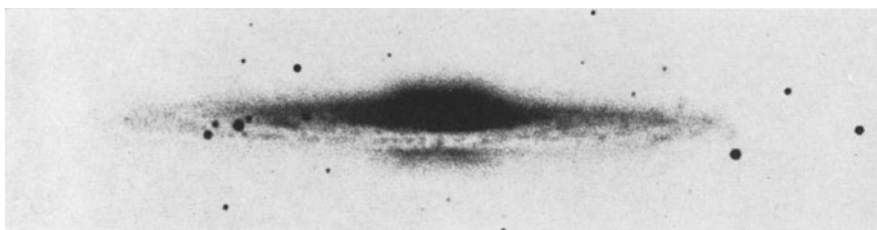
37. IC 1613 (P 48''): IBm . The residual structure includes the well resolved axial bar and asymmetrical bright patch rich in supergiants and emission objects. Note characteristic asymmetry. Compare with 36 and 38.



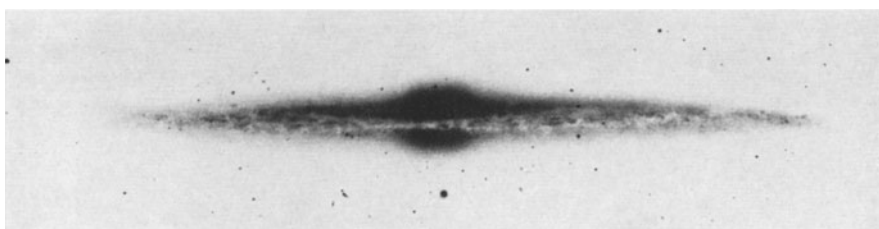
38. Dwarf I in Andromeda ($2^h31, +40^\circ19'$) (P 200''): $dIB(s)m$. The characteristic structure of the late-type $SB(s)$ spirals is still recognisable, including axial bar (foreshortened in the projection), with embryonic spiral arms emerging at its extremities; one of the arms can be faintly traced to the left and lower part of the field. Compare with 36 and 37.

Plate VIII. Edgewise systems: Ordinary spirals.

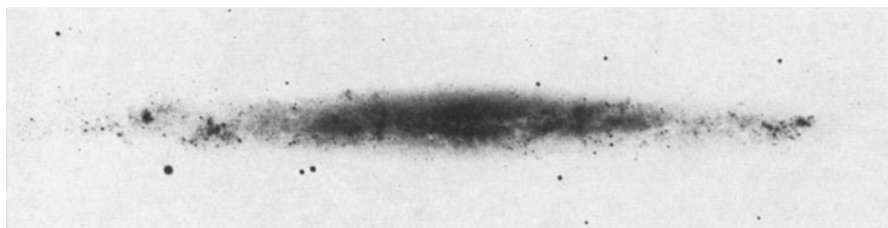
39. NGC 4594 (P 200''): $SA(s:)a$. The very bright and large spheroidal component, made up of high-velocity stars, emits most of the light of the object; the thin and weak flat component includes only very few bright Population I supergiants; it is surrounded by a heavy peripheral ring of dark matter; the spiral pattern is faint and smooth. Compare with 17 and 21.



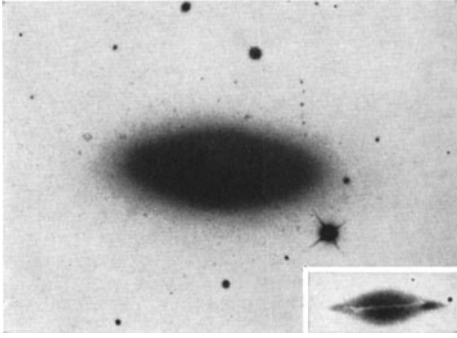
40. NGC 5746 (W 60''): $SA(s:)b$. The spheroidal component is very much reduced, but still contributes a large fraction of the light; the spiral pattern of bright and dark arms is well defined (save for the projection foreshortening), regular and still fairly smooth, although it begins to show some clumpiness. Compare with 18 and 22.



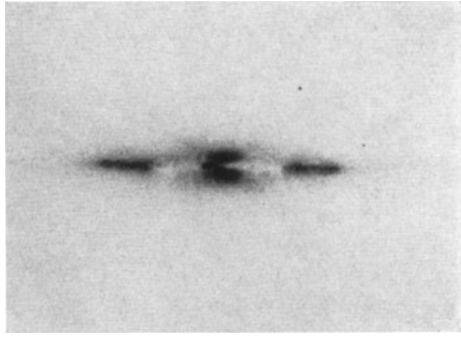
41. NGC 4565 (P 200''): $SA(s:)bc$. The still smaller spheroidal component is surrounded by an extensive flat component in which bright and dark spiral arms show much clumpiness and branching. Compare with 18, 19 and 23.



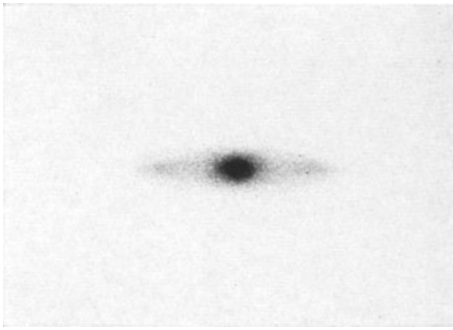
42. NGC 4244 (P 200''): $SA(s:)cd$. The spheroidal component is now inconspicuous in the center of a very extensive flat component in which the spiral pattern of bright and dark matter shows much irregularity and very many well resolved knots. Compare with 20 and 24.

Plate IX. Edgewise systems: Lenticulars and magellanic irregulars.

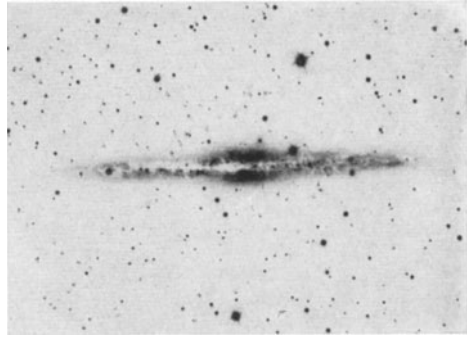
43. NGC 5866 (P 200''): $SA(s)O^{\circ}:(sp)$. The outer envelope simulates an elliptical, but the lens (inset) shows a narrow ring of dark matter surrounding the large nucleus. Compare with 3, 44 and 45.



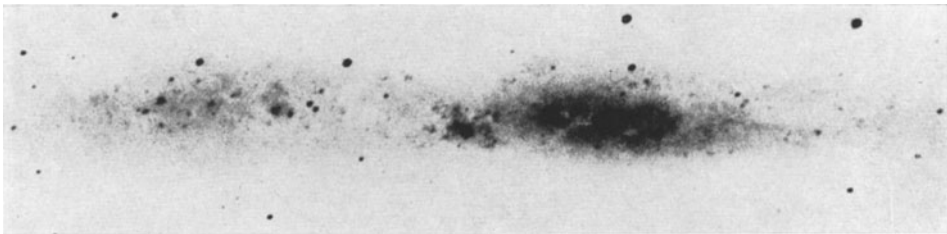
44. NGC 4710 (W 100''): $SA(rs?)O^{+}(sp)$. This short exposure shows only the lens and fairly small nucleus surrounded by a heavy ring of dark matter and bright ansae, being the edge-on view of an incomplete ring or incipient spiral arms.



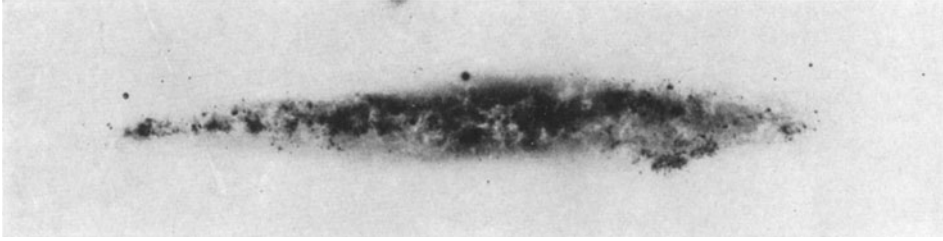
45. NGC 4215 (W 100''): $SA(r)O^{\circ}:(sp)$. This short exposure shows only the small nucleus and the weak ansae, being the edge-on view of the incipient ring structure in the lens. Compare with 7, 8 and 44.



46. NGC 891 (P 200''): $SA(s:)b(sp)$. This edge-on view of a typical ordinary spiral is for comparison with 43 and 44. Note how the dark matter, limited to a ring close to the nucleus in the lenticulars, spreads to the whole of the intermediate and outer parts of the spiral pattern. Compare also with 39 and 42.

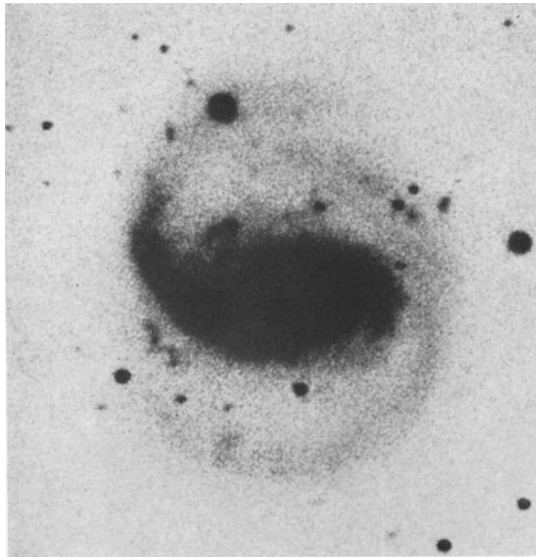


47. NGC 55 (S 30''): $SB(s)m(sp)$. The brighter part which is not resolved, except for a few bright emission knots and complex dark patches, represents a "rear" end-on view of the axial bar; the fainter part (left) which is partly resolved represents the end-on view of the main asymmetrical spiral arm. Between them dark clouds obscure the distant parts of the arm; weak embryonic arms can be identified to the right; compare with 36, 37, 38 and 48.

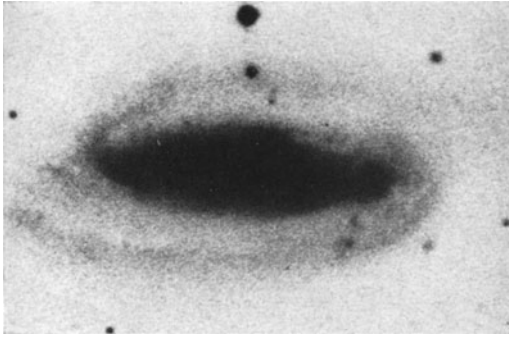


48. NGC 4631 (P 200''): $SB(s)d$ or $m(sp)$. The nearly central brighter part which is not resolved and against which complex dark clouds are projected represents a "front" end-on view of the axial bar; from it emerge two well-resolved, irregular spiral arms spreading out in opposite directions. Note characteristic asymmetry. Compare with 32, 38, 42 and 47.

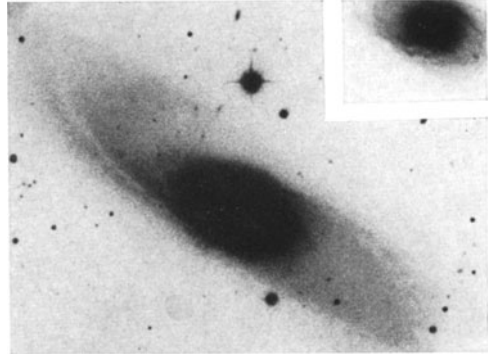
Plate X. Orientation effects on appearance of $SB(s)a$ or $SB(r)a$ spirals.



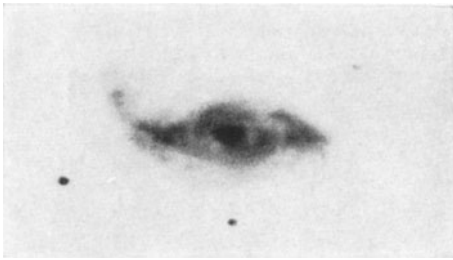
49. NGC 7552 (S 74''): $(R') SB(s)a$. Face-on view; same as 29; note outer loops. Details of bar and lens, seen from "above" are in 52; note pattern of dark matter.



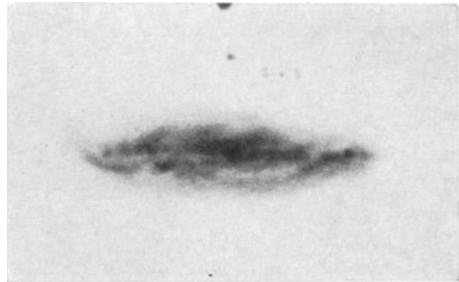
50. NGC 7582 (S 74'') : (R') $SB(s)a$. Broad-side view; details of bar and lens, seen from "across", are in 53; note pattern of dark matter.



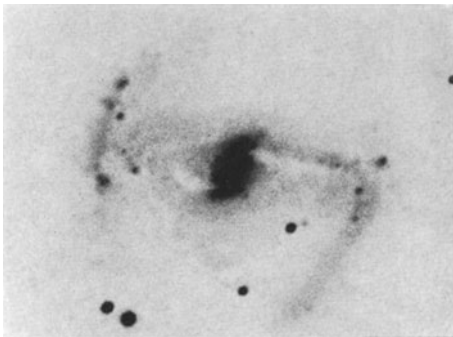
51. NGC 5566 (P 200'') : $SB(rs)a$. End-on view; details of bar and lens, seen from "along", are in inset. There is an (r) structure in the lens which does not show up on this print. Note lane or layer of dark matter splitting outer arms.



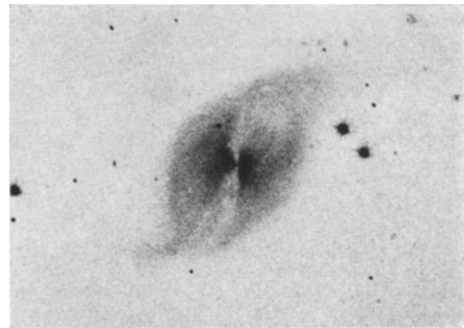
52. NGC 7552. Detail of bar and lens.



53. NGC 7582. Detail of bar and lens.



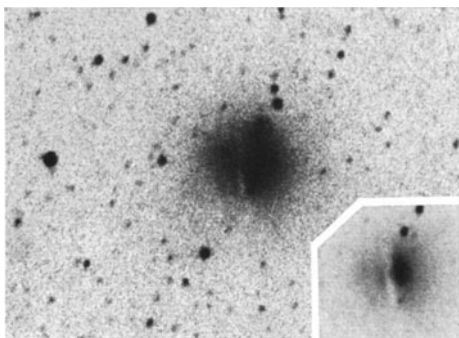
54. Lens of NGC 5383 (W 60'') : $SB(s)b$. Seen nearly "face-on" as in 49; note structure of nucleus and pattern of dark matter; compare with 30.



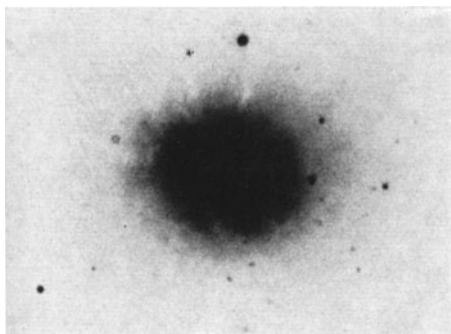
55. Lens of NGC 3718 (W 60'') : $SAB(s)a$. The bar is seen nearly "end-on" as in 51. Compare also with 30 and 54.

Very peculiar-looking structures can be produced in the lens and nucleus of early $SB(s)$ galaxies depending on tilt and orientation of the bar with respect to the line of sight.

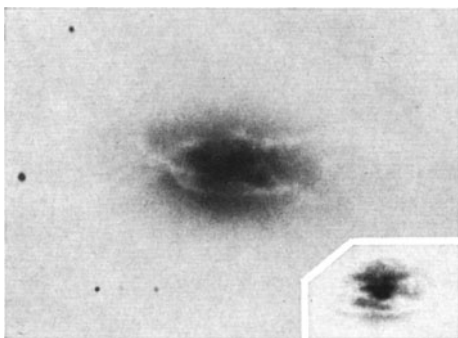
Plate XI. Peculiar lenticulars and non-magellanic irregulars.



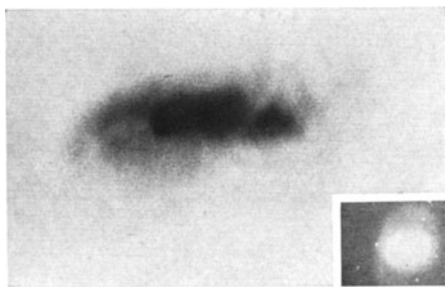
56. NGC 1947 (S 74''): *SAOp*. Little elongated, smooth nebulosity, rather similar to elliptical galaxy or to spheroidal bulge of early spiral, cut by two slightly curved lanes of dark matter on one side of nucleus. There is no trace of spindle or outer spiral structure.



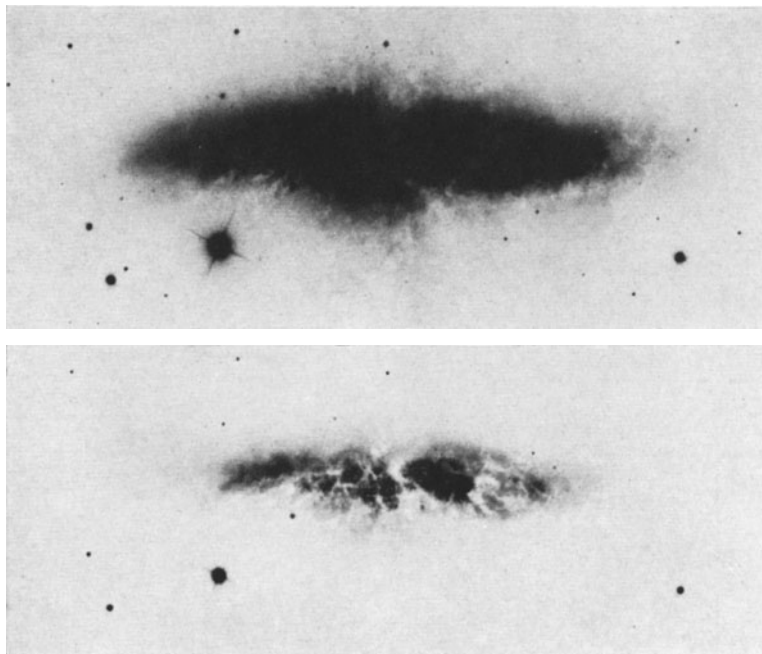
57. NGC 3077 (P 200''): *I(O)* or *SABOp*. Little elongated, smooth nebulosity marked by irregular patches of dark matter; two of them emerging on both sides near minor axis simulate embryonic spiral pattern as seen in nuclear regions of *SAB(s)O/a*. Has contrasting early spectral type and advanced colour index. Compare with 15.



58. NGC 4753 (W 100''): *I(O)* or *SB?* (*s*:) *O+p*. The inner isophotes (inset) are elongated at right angles to the direction of the major axis of the outer isophotes. The nuclear region is crossed by complex dark lanes which appear to be on the near side below the small nucleus and on the far side above it (see inset). This could be interpreted as the near end-on ("along") view of the bar and lens of an armless *SB(s)O/a* object. Compare with 55 and 56.



59. NGC 4691 (P 200''): (*R*) *SB(s:)O/a.p*. The weak outer whorls in the envelope (inset) show that this system is seen nearly face-on; note the complex structure of the bar crossed and surrounded by patches and curved lanes of dark matter. Compare with 52, 57, 58 and 60/61.



60/61. NGC 3034 = M 82 (P 200''): $I(O)$ or $SB(s)Op$. This is interpreted tentatively as the "broad-side", nearly edge-on view of the bar and lens of an armless $SB(s)O/a$ object. Has contrasting early-type spectrum and advanced colour index. Compare with 53, 57 and 59. Note especially the great similarity between 59 and 61 and the characteristic "crossing-over" of the main dark lane in the nucleus¹.

b) Quantitative morphology.

5. Apparent and true flattening. $\alpha)$ *Ellipticals*. The distribution of apparent and true flattening among elliptical galaxies has been investigated by HUBBLE [25], MACHIELS [45] to [47], TEN BRUGGENCATE [8], DE VAUCOULEURS [80] and others.

The elementary theory is simple, but its application to the physical situation is complicated by selection and other systematic effects and no more elaborate theory has yet been developed.

Consider an ellipsoid of revolution of equatorial diameter a_0 and polar axis b_0 whose equatorial plane is tilted at an (acute) angle i on the line of sight and which appears projected on the tangent plane to the celestial sphere as an ellipse of apparent axes $a = a_0$ and $b \leq b_0$, the true flattening is $e = 1 - (b_0/a_0) = 1 - q_0$, the apparent flattening is $\varepsilon = 1 - b/a = 1 - q$, and

$$\sin i = \sqrt{(q^2 - q_0^2)/(1 - q_0^2)} \quad (5.1)$$

For a random space orientation of the minor axes of the ellipsoids the probability that the inclination on the line of sight of the equatorial plane of any system lies between i and $i + di$, is $\cos i$. $di = d(\sin i)$, so that the probability

¹ This interpretation of the non-megallanic irregulars is consistent with their luminosities, colours and dimensions (cf. p. 315, Fig. 1 and 2). It also makes it possible to understand the rapid, solid-body rotation observed along the major axis of M 82 (N. U. MAYALL, in "Problems of Cosmical Aerodynamics", U.S. Central Air Documents Office, p. 181, 1951).

for an ellipsoid of true flattening e_m to appear of type En , i.e. with an apparent flattening ε_n , is

$$\Delta_{n-\frac{1}{2}}^{n+\frac{1}{2}}(\sin i) = \sqrt{\frac{q_{n+\frac{1}{2}}^2 - q_{n-\frac{1}{2}}^2}{1 - q_0^2}}. \quad (5.2)$$

HUBBLE [25] has computed for the eight groups of ellipticals $E0, E1, \dots, E7$ the theoretical frequencies of apparent flattening for each of the corresponding groups of true flattening. The limits of the flattening intervals chosen were 0.00–0.05–0.15–0.25–...–0.65–0.75; note that the range of flattening for the $E0$ groups (0.00–0.05) is only half as wide as for the other groups. This subdivision leads to the relative frequencies listed in Table 3.

Table 3. *Frequencies of apparent flattening among ellipsoids.*

Actual	Apparent								Total
	$E0$	$E1$	$E2$	$E3$	$E4$	$E5$	$E6$	$E7$	
$E7$. . .	0.055	0.111	0.114	0.116	0.121	0.132	0.164	0.187	1.000
$E6$. . .	0.059	0.123	0.126	0.133	0.148	0.187	0.224		1.000
$E5$. . .	0.067	0.140	0.148	0.166	0.216	0.263			1.000
$E4$. . .	0.079	0.169	0.190	0.250	0.312				1.000
$E3$. . .	0.100	0.225	0.299	0.376					1.000
$E2$. . .	0.145	0.378	0.477						1.000
$E1$. . .	0.300	0.700							1.000
$E0$. . .	1.000								1.000
Total . .	1.805	1.846	1.354	1.041	0.797	0.582	0.388	0.187	8.000
Frequency	0.226	0.231	0.169	0.130	0.100	0.073	0.049	0.023	1.000

The totals at the bottom of the table give the theoretical distribution of apparent flattening in a population of ellipsoids randomly oriented and equally distributed over the whole interval of e (0.00–0.75). HUBBLE used this table to analyse the observed distribution of 87 bright ellipticals among the eight groups $E0$ to $E7$; excluding 2 doubtful cases the relative frequencies for 85 objects were as follows:

$E0$	$E1$	$E2$	$E3$	$E4$	$E5$	$E6$	$E7$
20%	15%	17%	12%	15%	7%	8%	6%

By comparison with the last line of Table 3 there is an excess of strongly flattened shapes $E6$ to $E7$ and a deficiency of circular forms $E0$ to $E1$, but Hubble noted that much of the divergence could be due either to fluctuations in the small sample or to departures of the most flattened ellipticals from true ellipsoids and also to the inclusion in the $E7$ group of some unrecognized early-type spirals (actually lenticulars) seen edgewise.

The same data were again discussed by TEN BRUGGENCATE [8] and by MACHIELS [45] without significant modification of HUBBLE's conclusions. MACHIELS [47] and later WYATT¹ also analyzed the larger sample of rather more than 200 objects classified as E in the Shapley-Ames Catalogue and for which ε can be computed from the published dimensions; MACHIELS concluded that there is no statistically significant excess or deficiency of any group compared with a uniform distribution of true flattening², while WYATT is of the opinion that

¹ S. P. WYATT: *Astronom. J.* **55**, 187 (1950) (Abstract).

² See also discussion by MACHIELS [46] of the objects classified as "elliptical" in the Harvard Catalogue of the Coma-Virgo cluster (*Harvard Ann.* **88**, No. 1 (1930)) in which he emphasizes the errors introduced by the definition and unequal widths of the flattening intervals.

there is a significant excess of highly oblate systems over nearly spherical ones and further that there is a definite deficiency of systems of intermediate (true) flattening with a minimum near $e \approx 0.3$.

Because of the mixture of types in earlier statistics none of these conclusions can be accepted without further discussion and HUBBLE'S reasonable warning of possible systematic errors appears fully justified. In particular, it is now clear that the shapes of the isophotes of the most flattened "ellipticals" depart notably from similar geometrical ellipsoids (cf. p. 323, Fig. 7) and that many objects of the other classes, mainly lenticulars, have been erroneously included in the early statistics of ellipticals. As noted in Sect. 3 this is a most serious cause of confusion in the old data.

More significant results can probably be obtained from a discussion of 48 objects classified as E or E^+ in the Mt. Stromlo survey [80]. The observed relative frequencies in flattening intervals limited at $0.00-0.07-0.17-\dots-0.67-0.77$ are given in Table 4 for both the "estimated" types $E0$ (including $E0$ to 1), $E1$ (including $E1$ to 2), etc. and for the "measured" types derived from actual measurements of the axes of the photographic images¹.

Table 4. *Observed relative frequencies of estimated and measured flattening of ellipticals.*

Flattening ϵ	0.00	0.07	0.17	0.27	0.37	0.47	0.57	0.67	0.77	Total
Type	$E\ 0$	$E\ 1$	$E\ 2$	$E\ 3$	$E\ 4$	$E\ 5$	$E\ 6$	$E\ 7$		
Frequency of ϵ_{est}	23	17	17	13	16	8	4	2		100
Frequency of ϵ_{meas}	25	19	15	19	6	10	4	2		100
Mean	24	18	16	16	11	9	4	2		100

The mean frequencies agree closely with the theoretical values for a uniform distribution of true flattening and random orientation (Table 3). This is shown by the following summary of the relative frequencies observed and computed

Types	$E0-E1$	$E2-E3$	$E4-E5$	$E6-E7$
Observed (%)	42	32	20	6
Computed (%)	42	33	19	6

In view of the smallness of the sample on which this result rests, the conclusion must, however, remain provisional and subject to verification. A final discussion should be possible after the completion of the current revision of the Shapley-Ames galaxies for the whole sky.

β) *Lenticular galaxies.* The distribution of apparent and true flattening among lenticular galaxies as a separate class, distinct from the ellipticals, has not yet been studied in any detail. The distribution of apparent flattening ϵ_m in a sample of 33 lenticulars included in the Mt. Stromlo survey [80] is given in Table 5.

Table 5. *Frequency of apparent flattening among lenticulars.*

ϵ_m	0.0	0.2	0.4	0.6	0.8
Observed (%) .	31	27	12	30	
Random (%) . .	40	20	20	10	

¹ The mean systematic difference for the 48 objects is $\epsilon_m - \epsilon_e = +0.016$ and the average deviation (excluding nine $E0$ objects) is ± 0.034 . The internal probable error of the measured flattening is about ± 0.025 .

For comparison the last line gives the theoretical frequencies for a random distribution under the same assumptions as in Table 3. The deficiency in the interval 0.4 to 0.6 is probably accidental, but the deficiency of nearly circular objects ($\varepsilon < 0.2$) is probably real and the excess of elongated objects ($\varepsilon > 0.6$) is certainly significant. The observed distribution merely confirms that lenticulars are strongly flattened systems and behave in this respect more like spirals than ellipticals; a larger sample would be required for the derivation of the frequency distribution of true oblateness.

γ) *Spirals*. The problem of the true space orientation of the equatorial planes of spirals as derived from the distribution of apparent flattening has

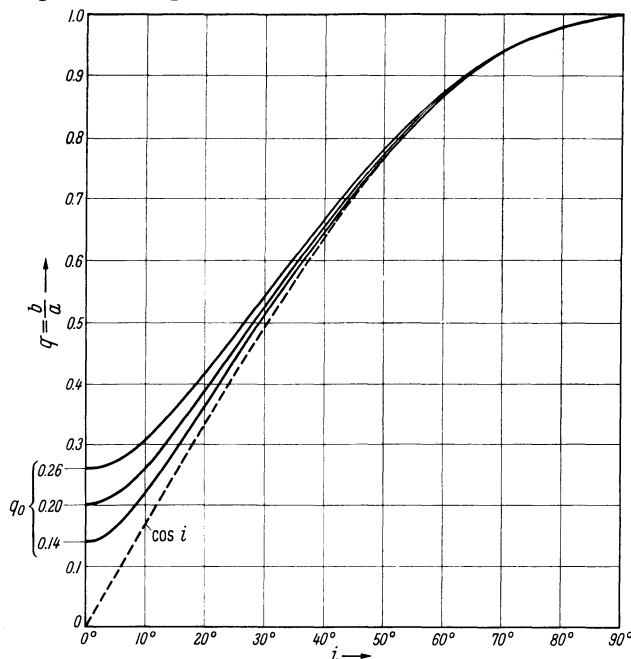


Fig. 5. Apparent flattening $q = b/a$ as a function of true flattening q_0 and tilt angle i for ellipsoids.

been the subject of many investigations with conflicting results. Some early studies had suggested that there is an excess of elongated shapes compared with expectations for a random (uniform) distribution of rotation axes¹; this has not been confirmed by other, more recent investigations², but the discussion is probably not yet closed³.

Consider a large number of flat circular discs whose axes are randomly oriented and which are observed from a distant point or one such disc observed from a large number of points uniformly distributed over the sphere; the probability for a given observer to be located between latitudes i and $i + di$ is proportional

to the area of the zone defined by these angles, i.e. $\cos i$. $di = d(\sin i)$. Hence a statistically uniform distribution of $\sin i$ characterizes a random orientation of the planes. For geometrical ellipsoids of apparent axial ratio $q = b/a$ and true axial ratio $q_0 = (b_0/a_0)$, $\sin i$ is given by (5.1). The relation between i and q is shown in Fig. 5 for a number of values of q_0 between 0.14 and 0.26. According to a discussion by HOLMBERG [21] the mean axial ratio of spirals seen exactly edge-on is $q_0 = 0.20$.

For the physical systems the situation is complicated by systematic errors and selection effects which have vitiated most—perhaps all—discussions. In

¹ e.g. J. H. REYNOLDS: Monthly Notices Roy. Astronom. Soc. London **81**, 129 (1920); **82**, 510 (1922). — F. G. BROWN: Monthly Notices Roy. Astronom. Soc. London **98**, 218 (1938); **99**, 14 (1938); see also [46].

² e.g. H. KNOX-SHAW: Monthly Notices Roy. Astronom. Soc. London **69**, 72 (1908); **98**, 587 (1938). — C. C. L. GREGORY: Monthly Notices Roy. Astronom. Soc. London **84**, 456 (1924); see also [12], [47].

³ e.g. E. ÖPIK: Observatory **46**, 51, 65 (1923). — R. O. REDMAN: Observatory **61**, 216 (1938). — F. G. BROWN: Observatory **61**, 250 (1938). — A. G. WALKER: Monthly Notices Roy. Astronom. Soc. London **100**, 623 (1940). — S. P. WYATT and F. G. BROWN: Astronom. J. **60**, 420 (1955).

the first place spiral galaxies depart strongly from ideal ellipsoids and the departure depends on the exact type, late-type spirals are more flattened than early-type objects; for instance the flat component including the spiral arms has a mean axial ratio of about 12:1 in *Sb* — *Sc* edgewise spirals¹ while the “spherical” component in the central bulge has a mean axial ratio of about 2 to 1 in *Sa* — *Sb* edgewise spirals ([12], p. 82 to 86); then the ratio between the spherical component and the flat component varies greatly along the spiral sequence. Another difficulty is the presence in the measured axial ratios of systematic errors which have been thoroughly investigated experimentally by HOLMBERG [21]; in general the flattening derived from micrometric measurements of the plates is exaggerated as compared with photometric determinations of the isophotes; such measuring errors must be corrected before any conclusion can be derived concerning the distribution of true and apparent flattening. A third effect is the projection or tilt effect: a given galaxy will be assigned a larger diameter seen edgewise than when seen face-on; for a transparent oblate spheroidal system of true axial ratio q_0 and projected ratio q , in which the radial intensity distribution in the face-on view is $I_1(x, 0)$, it can be shown² that the intensity distribution along the major axis of the projected image is given by

$$I_i(x, 0)/I_1(x, 0) = 1/q = a/b \geq 1 \quad (5.3)$$

in particular, for the edge-on view ($i = 0^\circ$)

$$I_0/I_1 = 1/q_0 > 1.$$

Hence, in a sample selected according to apparent diameter, edgewise systems will be included that would not have been included had they been oriented face-on; this introduces a spurious excess of elongated shapes compared with a sample from a given volume of space. The exact amount of this selection effect cannot be computed theoretically and it is very difficult to determine empirically.

From a discussion of 270 spirals north of $\delta = -20^\circ$ listed in REINMUTH's catalogue and whose major diameters are not less than 3'0, after correction for measuring errors and projection effects, HOLMBERG [21] concluded that the distribution of $\sin i$ is quasi uniform and therefore that the axes are randomly oriented. A similar study of 156 spirals larger than 3'0 listed in REIZ' catalogue led to the same conclusion. However, the basic material (the Heidelberg Bruce collection) is common to both catalogues and the correction for projection effect was more or less arbitrarily selected so as to produce the desired result. Furthermore, the basic statistics of the frequency distribution of apparent axial ratio among spirals are in serious disagreement as shown in Table 6 and Fig. 6 where

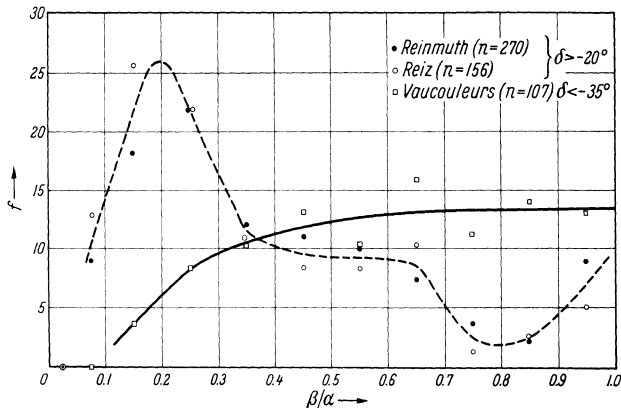


Fig. 6. Frequency distribution of apparent axial ratio among spirals in the Heidelberg and Mount Stromlo surveys.

¹ A. B. WYSE and N. U. MAYALL: *Astrophys. Journ.* **95**, 39 (1942).

² S. P. WYATT and F. G. BROWN: *Astronom. J.* **60**, 420 (1955).

the two sets of Heidelberg data (uncorrected and corrected for measuring errors only)¹ are listed.

Table 6. *Frequency of axial ratio and tilt among spirals.*
Absolute frequency of apparent axial ratio (uncorrected).

β/α	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0.05	Sum
REINMUTH	24	6	10	20	27	30	33	59	49	12		270
REIZ	8	4	2	16	13	13	16	30	40	10		156
VAUCOULEURS	14	15	12	17	11	14	11	9	4	0		107

Relative frequency of tilt angle (corrected for measuring errors).

$\sin i$	1.0	0.8	0.6	0.4	0.2	0.0	Remarks
REINMUTH	0.16 ₅	0.15	0.21 ₅	0.23 ₅	0.24 ₅		$\bar{q} = 0.20 \quad a \geq 3'0$ $q = 0.14 \quad m_H \leq 13.0$
REIZ	0.13	0.20	0.18 ₅	0.20 ₅	0.28		
VAUCOULEURS	0.27	0.26	0.21 ₅	0.18	0.07 ₅		

The Heidelberg data indicate a conspicuous deficiency of nearly round shapes ($0.7 < \beta/\alpha < 0.9$) and a large excess of elongated shapes ($0.2 < \beta/\alpha < 0.4$), while the Mt. Stromlo data indicate a nearly constant distribution of flattening for $b/a > 0.4$ and a gradual deficiency of elongated shapes for $b/a < 0.4$ (comparison with other data indicates that measuring errors are negligible in the Mt. Stromlo data). The selection rule for the Heidelberg data is $a \geq 3'0$, while it is $m_{\text{Harvard}} \leq 13.0 \pm$ for the Mt. Stromlo data; this, however, does not account for the discrepancy since a segregation of the Mt. Stromlo data according to corrected diameter (reduced to face-on) indicates a still greater excess of round shapes for the larger objects².

A possible reason for the excess of elongated shapes in the Heidelberg data is the mixture of types and in particular the inclusion of edgewise lenticulars with spirals; as noted by HOLMBERG [21] the deficiency of nearly round shapes is probably partly subjective (e.g. decimal error). The statistical treatment may account for the deficiency of elongated shapes in the Mt. Stromlo data. A final discussion will be possible when the revised types and dimensions of the brighter spirals over the whole sky are published.

6. Shape of spiral arms. The shape of spiral arms has been studied by VON DER PAHLEN³, GROOT⁴, REYNOLDS⁵ and in greater detail by DANVER [12] who has determined the parameters of the best fitting logarithmic spiral for 98 systems.

The shape of the arms in the equatorial plane of the spiral can be found empirically by projecting a photograph on an inclined plane whose tilt and orientation are adjusted by trial and error until the projected outline appears roughly circular. It can be also determined analytically.

¹ For the observers who measured the Heidelberg Bruce Collection the mean relation between measured (β/α) and corrected (b/a) axial ratio is as follows:

b/a	1.00	0.8	0.6	0.4	0.2
β/α	1.00	0.76	0.53	0.31	0.125
$\beta/\alpha : b/a$	1.00	0.95	0.88	0.77	0.63

² Median axial ratio: 0.72 for $D > 4'0$; 0.63 for $3'0 < D \leq 4'0$; 0.47 for $2'0 < D \leq 3'0$; for a random distribution Fig. 5 shows that the median ratio is in the range 0.52 to 0.55.

³ H. VON DER PAHLEN: *Astronom. Nachr.* **188**, 249 (1911).

⁴ H. GROOT: *Monthly Notices Roy. Astronom. Soc. London* **85**, 535—541 (1925); **86**, 146 (1926).

⁵ J. H. REYNOLDS: *Monthly Notices Roy. Astronom. Soc. London* **85**, 1014—1020 (1925).

Let r, φ be the polar coordinates of a point P of an arm in the plane of the spiral and ϱ, ϑ its coordinates in the tangent plane to the celestial sphere, if $\omega = \frac{\pi}{2} - i$ is the angle between the plane of the spiral and the tangent plane, then

$$r = \varrho \sqrt{1 + \tan^2 \omega \cdot \sin^2 \vartheta} = \varrho \cdot c, \quad (6.1)$$

$$\tan \varphi = \tan \vartheta / \cos \omega. \quad (6.2)$$

$c = 1$ for $\vartheta = 0^\circ$ and 180° , $c = \sqrt{1 + \tan^2 \omega}$ for $\vartheta = 90^\circ$ and 270° . If the spiral is logarithmic, as suggested by the empirical results,

$$\text{Log } r = A - B\varphi \quad (6.3)$$

the measured values of $r = \varrho$ for $\vartheta = 0^\circ$ and 180° give A and B , then r is computed and ϱ measured for $\vartheta = 90^\circ$ and 270° , whence $\cos \omega = \varrho/r$. If circular symmetry is assumed in the plane of the spiral for the outer outline of the system another determination is $\cos \omega = b/a$, neglecting the small thickness of the flat component.

In practice the empirical method is quicker and just as accurate; for moderately tilted spirals ω can be determined to within 2° or 3° according to DANVER [12]. Usually the best fit is given by a logarithmic spiral, if

$$r = r_0 \cdot \exp(\kappa \alpha) \quad \text{with} \quad \alpha = \pi \varphi / 180^\circ, \quad (6.4)$$

$$\text{Log } r = \text{Log } r_0 + \kappa c \varphi \quad \text{where} \quad c = \frac{\pi}{180} \text{Log } e = 0.00758 \quad (6.5)$$

and

$$\kappa = \cot \mu, \quad (6.6)$$

μ being the characteristic angle between the tangent and the radius vector. The best fitting spiral is obtained through a least-square solution in $x = \varphi$ and $y = \log r$ coordinates which gives $\log r_0$ and κc . The origin $\varphi = 0$ may be taken at the visible extremity of an arm. A plot of $\text{Log } r$ vs. φ for the two main arms of NGC 4303, type $SAB(rs)bc$, measured by DANVER on a Lick photograph is shown in Fig. 7. The mean characteristic angle for 190 arms in 98 objects is $\bar{\mu} = 73.4 \pm 0.4$ (p.e.) with a dispersion $\sigma = 6.1 \pm 0.3$ (p.e.); the range of individual values is from $\mu = 54^\circ$ to $\mu = 86^\circ$. Within the same object μ is not the same for different arms and the corresponding dispersion is $\sigma = 4.9 \pm 0.3$. Along a given spiral arm there is a slight tendency for μ to increase outwards, at least in the outer parts. The mean values of μ decrease along the morphological sequence from Sa to Sc , but the change is only slight between Sb and Sc (Table 7).

Table 7. Mean geometrical parameters of spiral arms.

Type	SAb	SAc	SbB	SbC	All
μ	75.2 ± 0.9	72.5 ± 0.6	75.2 ± 1.0	73.0 ± 2.3	73.4 ± 0.4
σ_μ	5.9 ± 0.7	6.4 ± 0.4	5.0 ± 0.7	8.3 ± 1.6	6.1 ± 0.3
n	41	112	24	13	190
$\bar{L}$	1.80 ± 0.09	1.74 ± 0.05	1.62 ± 0.08	1.61 ± 0.09	1.76 ± 0.04
σ_L	0.58 ± 0.06	0.51 ± 0.03	0.38 ± 0.05	0.34 ± 0.06	0.54 ± 0.03
W	$328^\circ \pm 20^\circ$	$317^\circ \pm 10^\circ$	$277^\circ \pm 17^\circ$	$211^\circ \pm 24^\circ$	$310^\circ \pm 11^\circ$
σ_W	$125^\circ \pm 14^\circ$	$105^\circ \pm 7^\circ$	$82^\circ \pm 12^\circ$	$88^\circ \pm 17^\circ$	$115^\circ \pm 6^\circ$

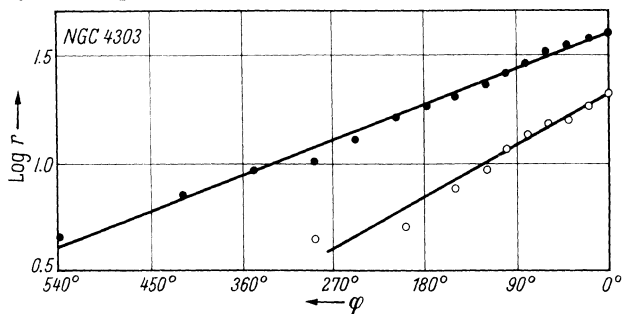


Fig. 7. Comparison of spiral arms of NGC 4303 with logarithmic spirals, after C. G. DANVER.

The angular winding W of the arms may be measured by the value of φ for which they apparently reach the nucleus; their linear development L by the ratio of their total length to the apparent major axis a . The mean angular winding for 190 arms is $W = 310^\circ \pm 11^\circ$ (p.e.), i.e. the arms can be followed over slightly less than one complete turn, with a large dispersion $\sigma_W = 115^\circ \pm 6^\circ$ (p.e.); their average length is $L = 1.76 \pm 0.04$, with $\sigma_L = 0.54 \pm 0.03$; one arm is usually longer than the other, on the average $\Delta W = 58^\circ \pm 5^\circ$, $\Delta L = 0.45 \pm 0.05$. The length of the arms, measured in both W and L decreases along the sequence, late-type spirals having shorter arms (Table 7). There is a slight correlation between apparent diameter a and W , $r_{a,W} = +0.32 \pm 0.07$, probably indicating the presence of some systematic errors in the measurement of W as the arms are more easily traced in the larger objects ($W \approx 250^\circ$ for $a < 3'$, $W \approx 420^\circ$ for $a > 21'$); there is, however, no perceptible correlation between a and L .

Note that in any study of the geometrical shapes of spiral arms there is a strong selection of the more regular and simple systems; in many objects whose spiral structure has a multiplicity greater than 3 or 4 it is practically impossible to follow individual whorls for more than a small fraction of a turn. Further, in many systems the spiral arms are made up of a large number of shorter arcs more strongly inclined on the radius vector (i.e. having smaller μ), than the overall pattern. Such structure is also observed in some ringed types for which part of the structure $\mu \approx 90^\circ$. Finally in many barred spirals the faint extensions of the arms seem to return toward the bar and follow a general path more like a circle or an ellipse than a logarithmic spiral (cf. Sect. 4).

Index of illustrated galaxies.

NGC	Type	Plate and Figure	NGC	Type	Plate and Figure
45	<i>SA(s) dm</i>	VII, 34	4594	<i>SA(s:) a</i>	VIII, 39
55	<i>SB(s) m</i>	IX, 47	4631	<i>SB(s) dm</i>	IX, 48
300	<i>SA(s) d</i>	VII, 33	4691	<i>(R)SB(s:) O/ap</i>	XI, 59
488	<i>SA(r) b</i>	II	4710	<i>SA(rs?) O⁺</i>	IX, 44
891	<i>SA(s:) b</i>	IX, 46	4736	<i>(R)SA(r) ab</i>	V, 21
1084	<i>SA(s) c</i>	V, 19	4753	<i>I(O)</i>	XI, 58
1291	<i>(R)SB(s) O⁺</i>	IV, 12	5194	<i>SA(rs) bc</i>	II
1313	<i>SB(s) d</i>	VI, 32	5204	<i>SA(s) m</i>	VII, 35
1365	<i>(R')SB(s) b</i>	II; VI, 30	5273	<i>SA(s) O⁻</i>	III, 2
1433	<i>(R')SB(r) a</i>	II; VI, 25	5383	<i>SB(s) b</i>	X, 54
1512	<i>SB(r) O⁺</i>	IV, 14	5566	<i>SB(rs) a</i>	X, 51
1553	<i>SA(r) O^o</i>	III, 7	5746	<i>SA(s:) b</i>	VIII, 40
1566	<i>SAB(s) b</i>	II	5866	<i>SA(s) O^o</i>	IX, 43
1617	<i>SB(s) O⁺</i>	IV, 10	5962	<i>SA(r) c</i>	V, 23
1947	<i>SAOp</i>	XI, 56	6643	<i>SA(e) d</i>	V, 24
2523	<i>SB(r) bc</i>	VI, 27	6744	<i>SAB(r) bc</i>	II
2855	<i>SA(s) O⁺</i>	III, 4	6753	<i>(R)SA(r) b</i>	V, 22
2859	<i>(R)SB(r) O^o</i>	IV, 16	6873	<i>SAB(rs) O^o</i>	IV, 11
3031	<i>SA(s) ab</i>	II	7079	<i>SB(s) O^o</i>	IV, 9
3032	<i>SAB(r) O^o</i>	IV, 15	7124	<i>SB(rs) b</i>	II
3034	<i>I(O)</i>	XI, 60/61	7144	<i>EO</i>	III, 1
3077	<i>I(O)</i>	XI, 57	7166	<i>SA(r) O⁻</i>	III, 6
3351	<i>(R')SB(r) ab</i>	VI, 26	7205	<i>SA(s) bc</i>	V, 18
3367	<i>SB(r) cd</i>	VI, 28	7457	<i>E⁺ 5</i>	III, 5
3718	<i>SAB(s) a</i>	X, 55	7552	<i>(R')SB(s) a</i>	VI, 29; X, 49/52
4215	<i>SA(r) O^o:</i>	IX, 45	7582	<i>(R')SB(s) a</i>	X, 50/53
4244	<i>SA(s) cd</i>	VIII, 42	7702	<i>SA(r) O⁺</i>	III, 8
4262	<i>SB(r) O^o</i>	IV, 13	7741	<i>SB(s) cd</i>	VI, 31
4303	<i>SAB(rs) bc</i>	II	7793	<i>SA(s) d</i>	V, 20
4459	<i>SA(s) O^o</i>	III, 3	SMC	<i>SB(s) mp</i>	VII, 36
4565	<i>SA(s:) bc</i>	VIII, 41	I 1613	<i>IBm</i>	VII, 37
4569	<i>SA(s) ab</i>	V, 17	And I	<i>dIB(s) m</i>	VII, 38